Controller Area Network (CAN)

By:

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Session Objectives

- To identify the need for Automotive Communication Protocols
- To understand the characteristics of CAN protocol
- To study the working principles of CAN protocol

Session Topics

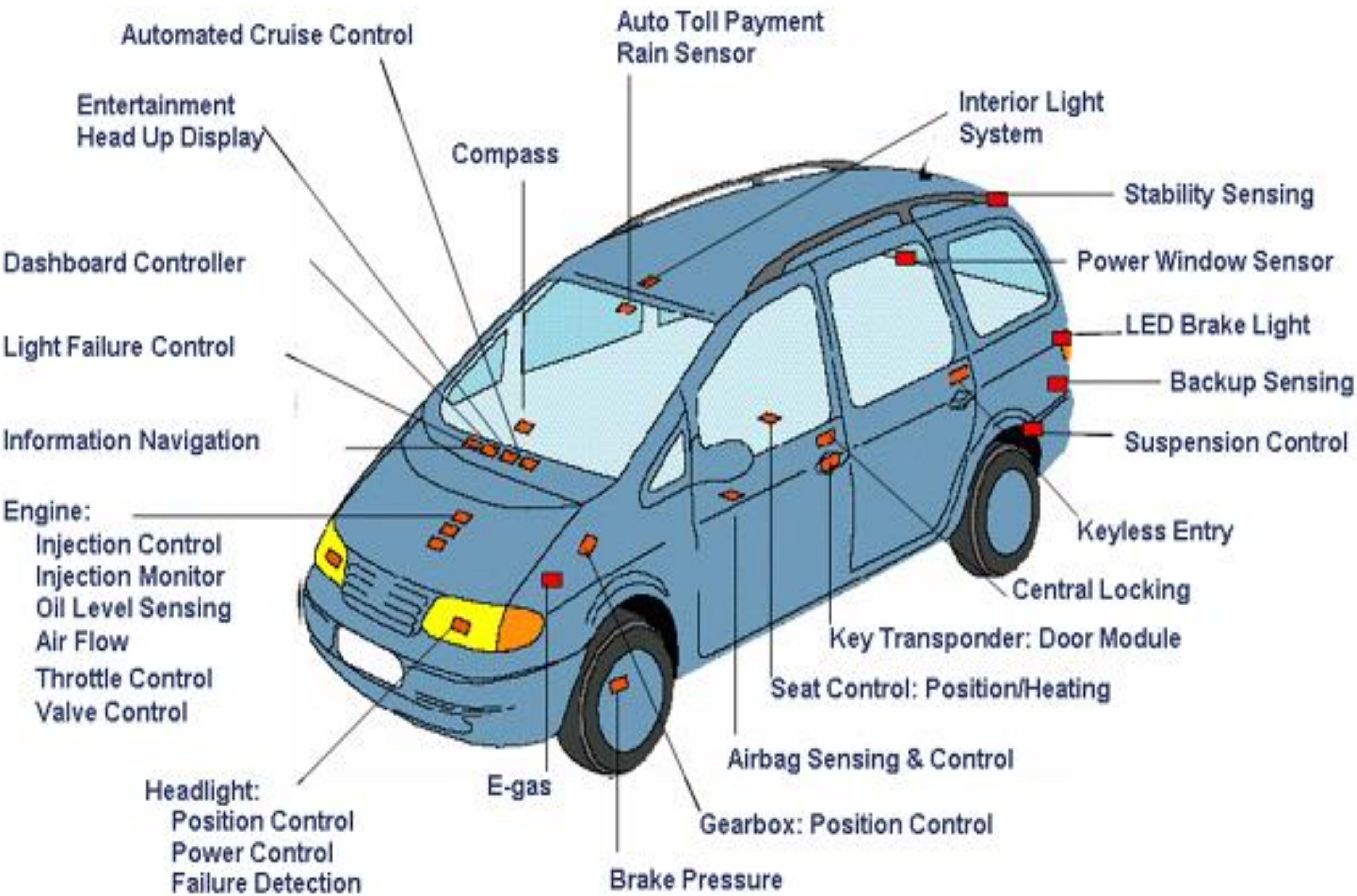
- Introduction to In-Vehicle Networking
- Classification of vehicle busses
- Vehicle multiplexing
- Evolution of CAN
- Characteristics of CAN protocol
- CAN in ISO-OSI reference model
- CAN frame format
- Error processing in CAN
- CAN physical layer
- Bit time and synchronization
- CAN physical media

Introduction

- Exponential increase in the number and sophistication of electronic systems in vehicles in past few decades

- Demands on power and design have led to innovations in electronic networks for automobiles

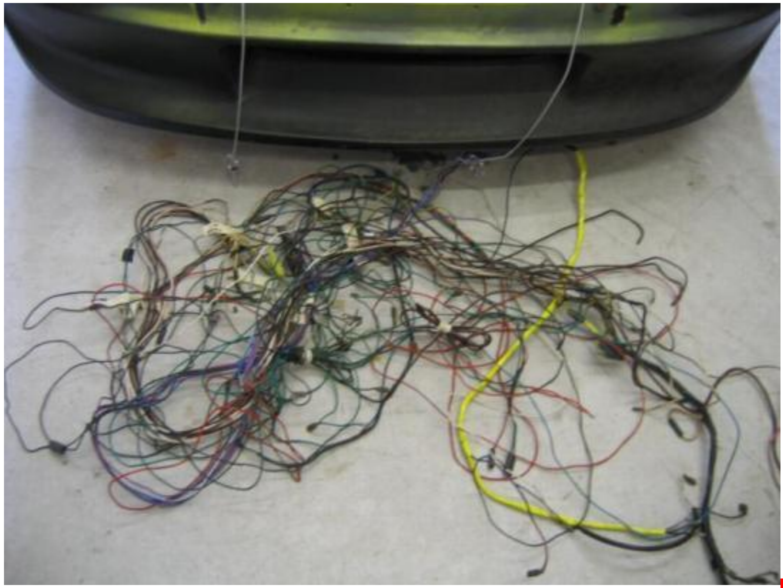
Example Components to be Networked

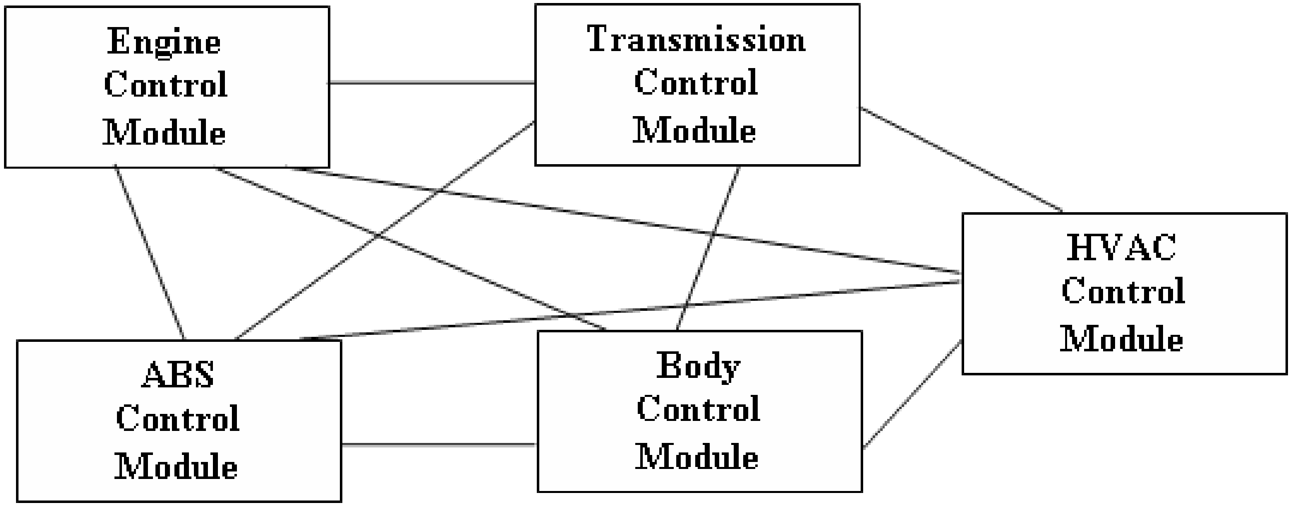


Need for In-Vehicle Networking

- To achieve communication between circuits, sensors, and many other electrical components within vehicles

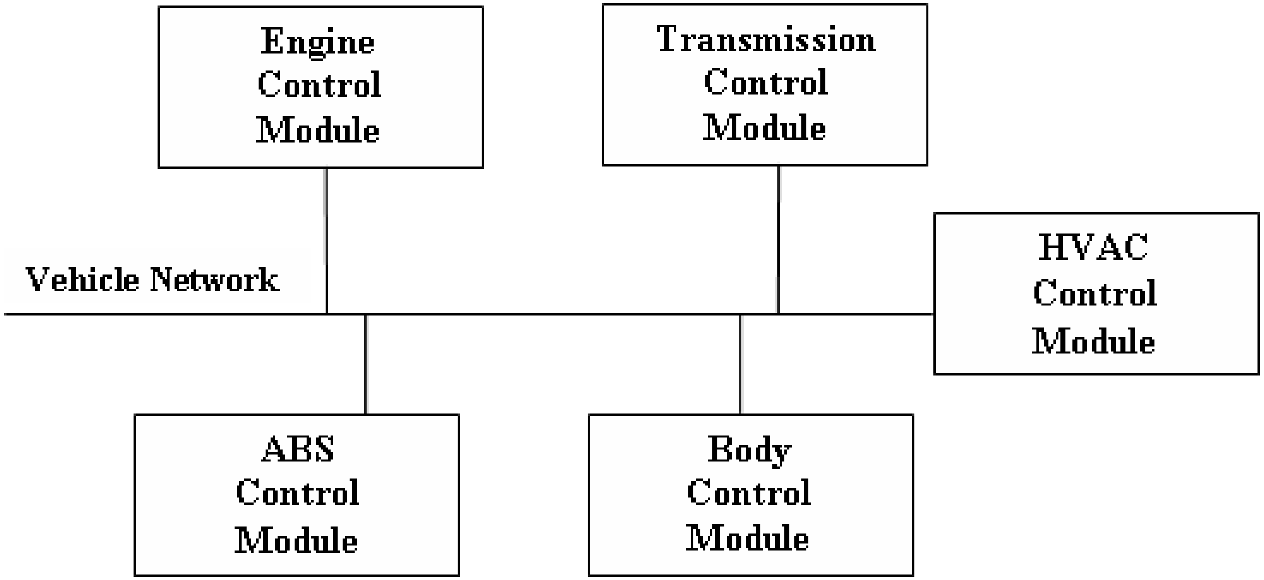
- Dedicated inter-module communication requires that requires point-to-point wiring is:
  - Bulky,
  - Expensive,
  - complex, and
  - difficult to install wiring harnesses



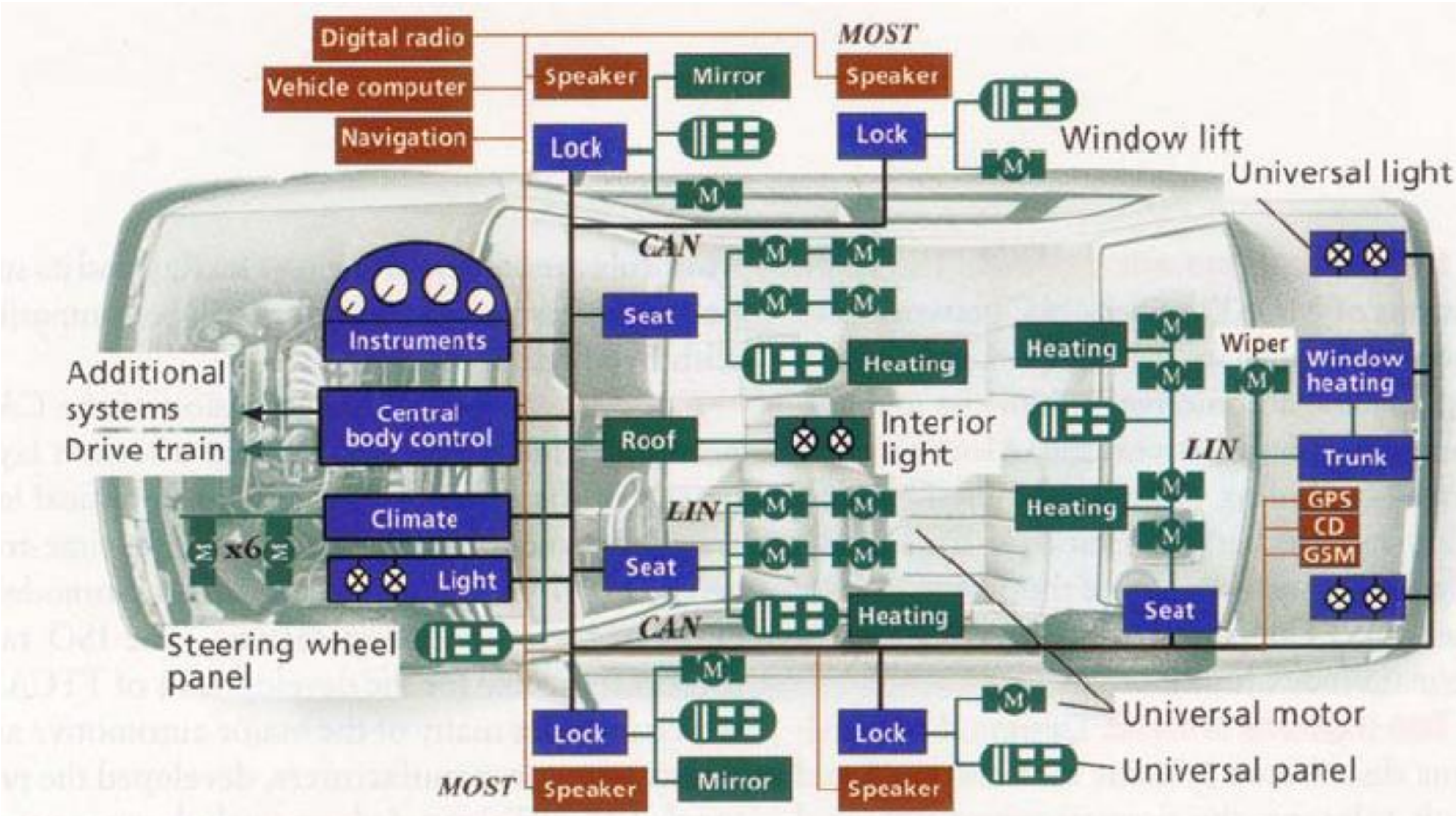


Point-to-Point Connections

Serial Bus



Example of In-Vehicle Network



CAN Controller area network
GPS Global Positioning System
GSM Global System for Mobile Communications
LIN Local interconnect network
MOST Media-oriented systems transport

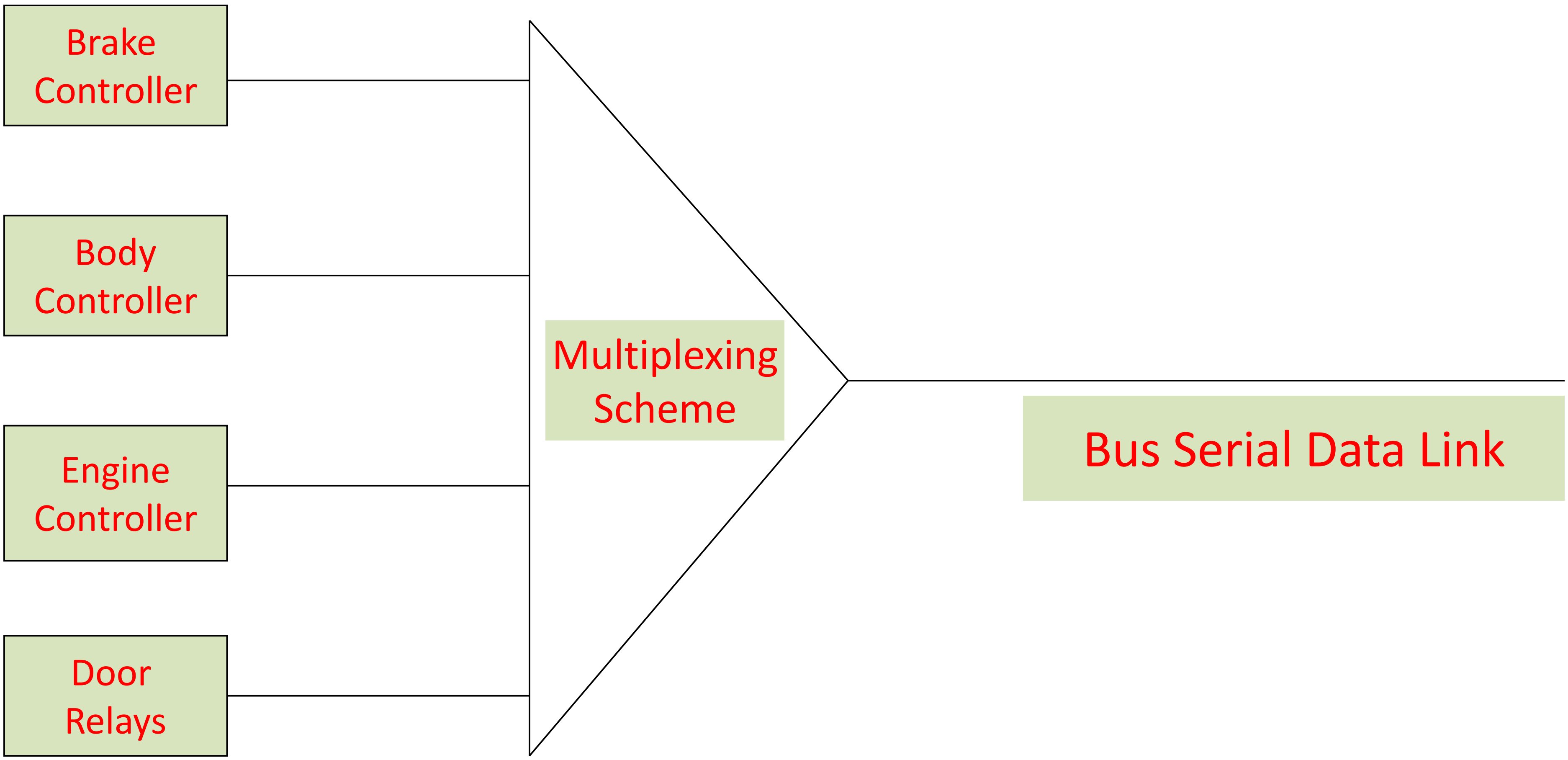
Classifications of Vehicle Buses

- Class A
  - Power windows, door locks, power seats, mirror control, wiper
  - Required data rates up to 125 kbps
  - Local Interconnect Network (LIN)
- Class B
  - Cruise control, HVAC, drive train, tire pressure
  - Required data rates up to 1 Mbps
  - Controller Area Network (CAN)
- Class C
  - Brake control, ABS, engine control, suspension control
  - Real-time operation, required data rates of up to 10 Mbps
  - Flexray

Classifications of Vehicle Buses

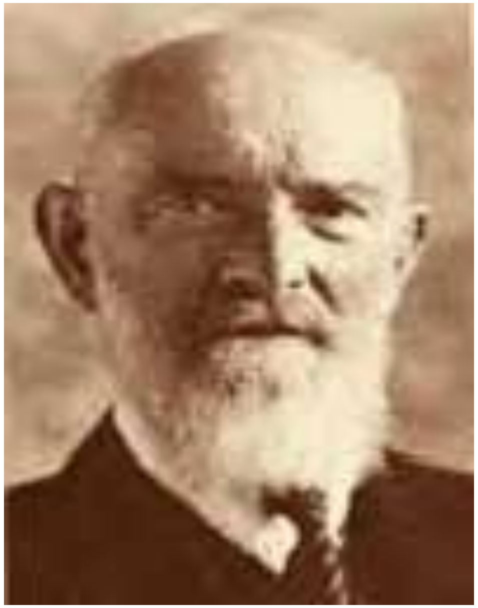
- Class D
  - Multimedia applications, navigation systems, audio, video
  - Required data rates of 10 Mbps or faster
  - Media Oriented System Transport (MOST)
- Emissions/Diagnostics
- Mobile Media
- X-by-Wire

Vehicle Multiplexing



Controller Area Network (CAN)

Foundation of CAN



- Created by German automotive system supplier Robert Bosch in the mid-1980s for automotive applications as a method for enabling robust serial communication

- The goal was to make automobiles more reliable, safe and fuel-efficient while decreasing wiring harness weight and complexity

Evolution

- 1983
  - Start of the Bosch internal project to develop an in-vehicle network
- 1986
  - CAN developed by Robert Bosch GmbH, Germany, for communication between three ECUs is vehicles designed by Mercedes
- 1987
  - First CAN silicon fabricated by Intel and Philips Semiconductors
- 1991
  - Bosch's CAN specification 2.0 published
  - CAN Kingdom CAN-based higher-layer protocol introduced by Kvaser

Evolution

- 1992
  - CAN in Automation (CiA) international users and manufacturers group established
  - CAN Application Layer (CAL) protocol published by CiA
  - First cars from Mercedes-Benz used CAN network

- 1993
  - ISO 11898 standard published

- 1994
  - 1st international CAN Conference (iCC) organized by CiA
  - DeviceNet protocol introduction by Allen-Bradley

Evolution

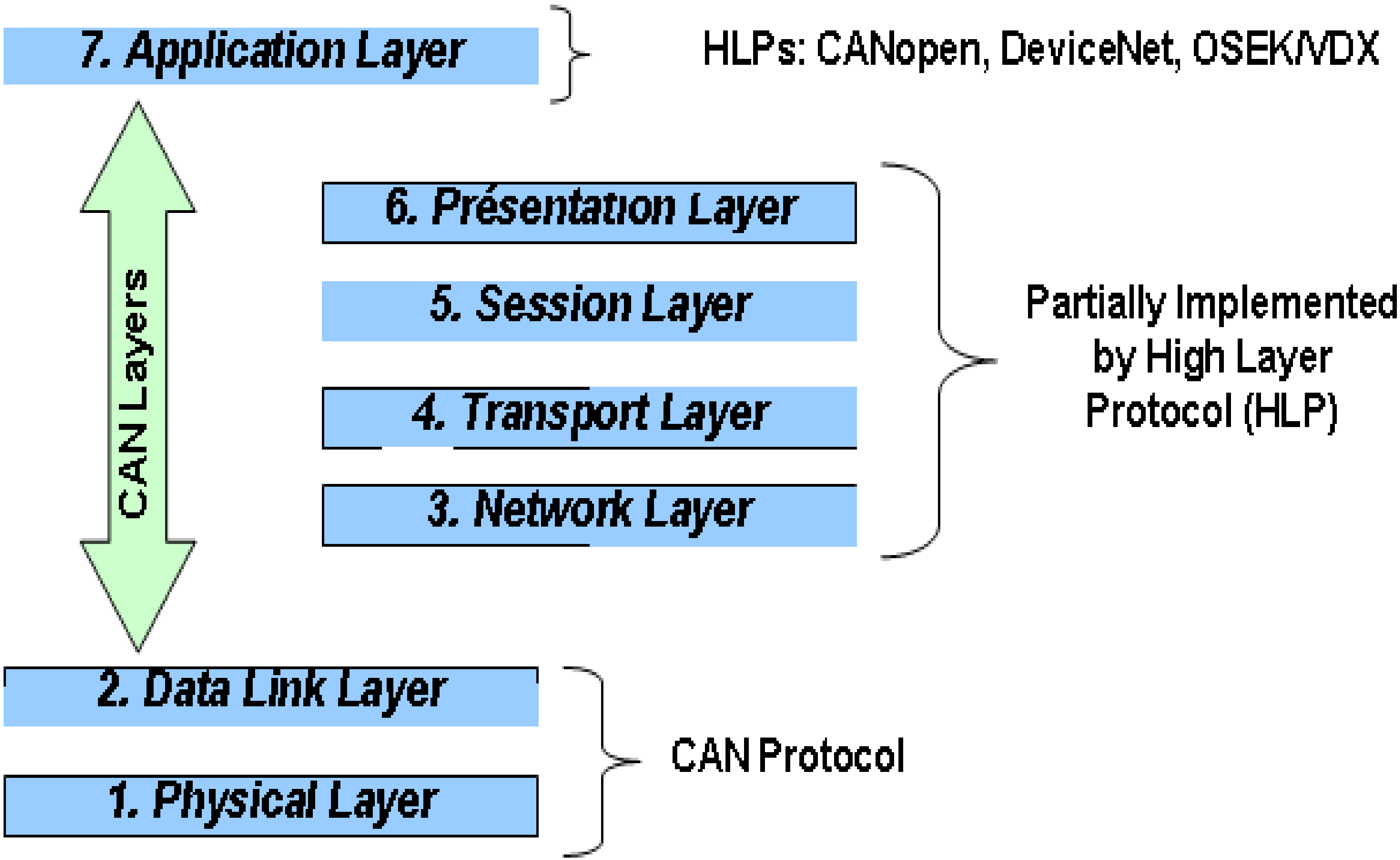
- 1995
  - ISO 11898 amendment (extended frame format) published
  - CANopen protocol published by CiA

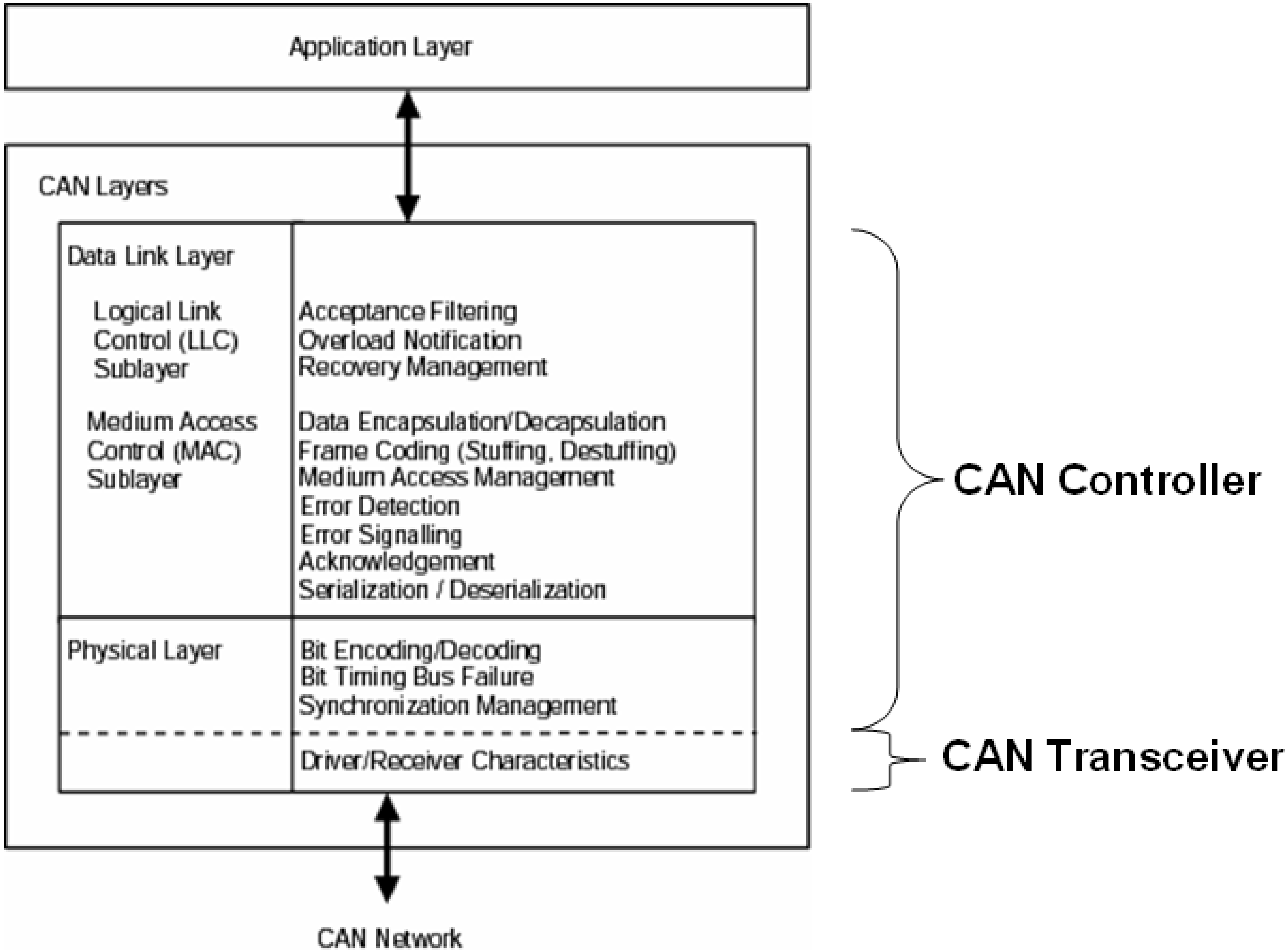
- 2000
  - Development of the time-triggered communication protocol for CAN (TTCAN)

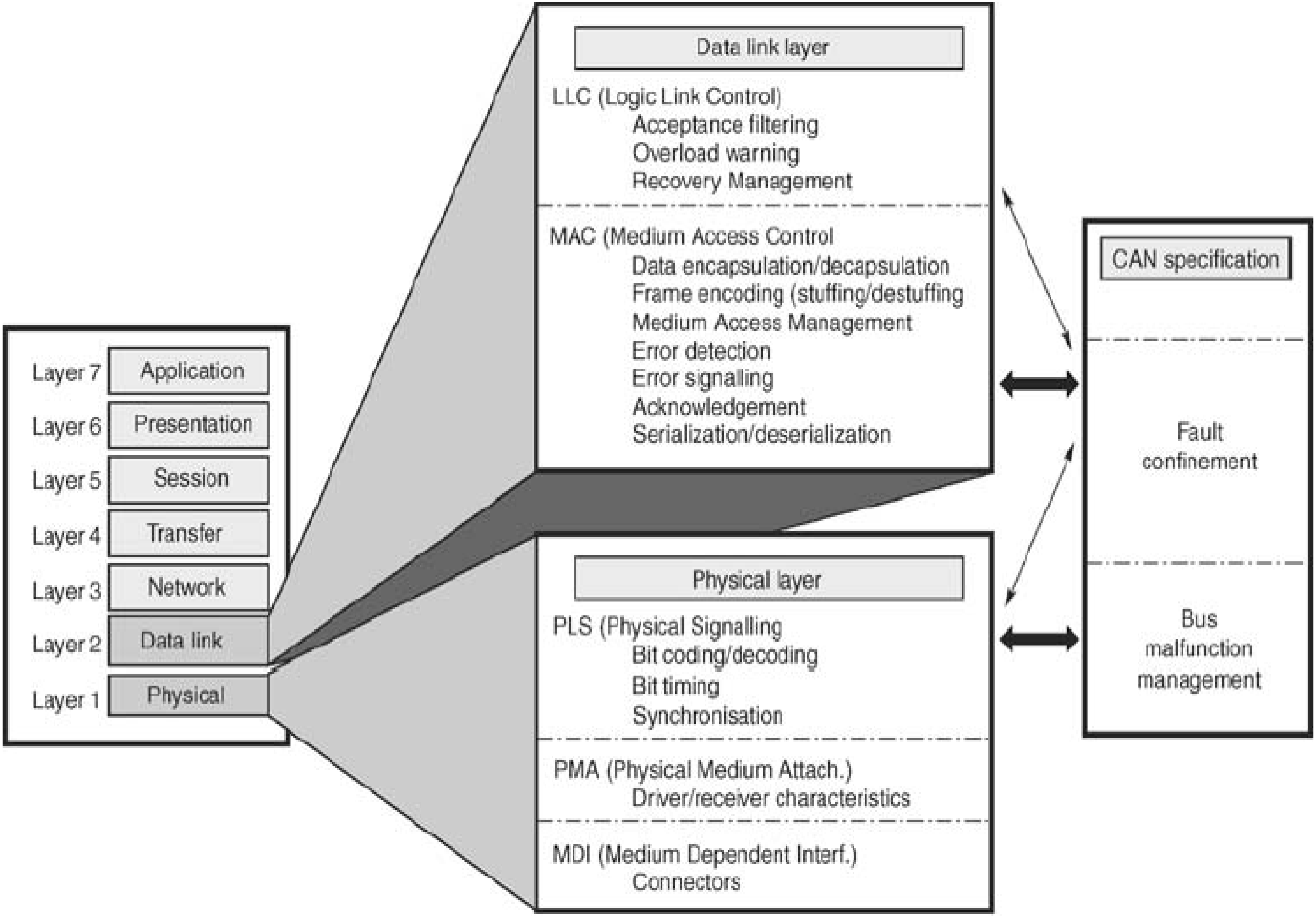
Characteristics of CAN Protocol

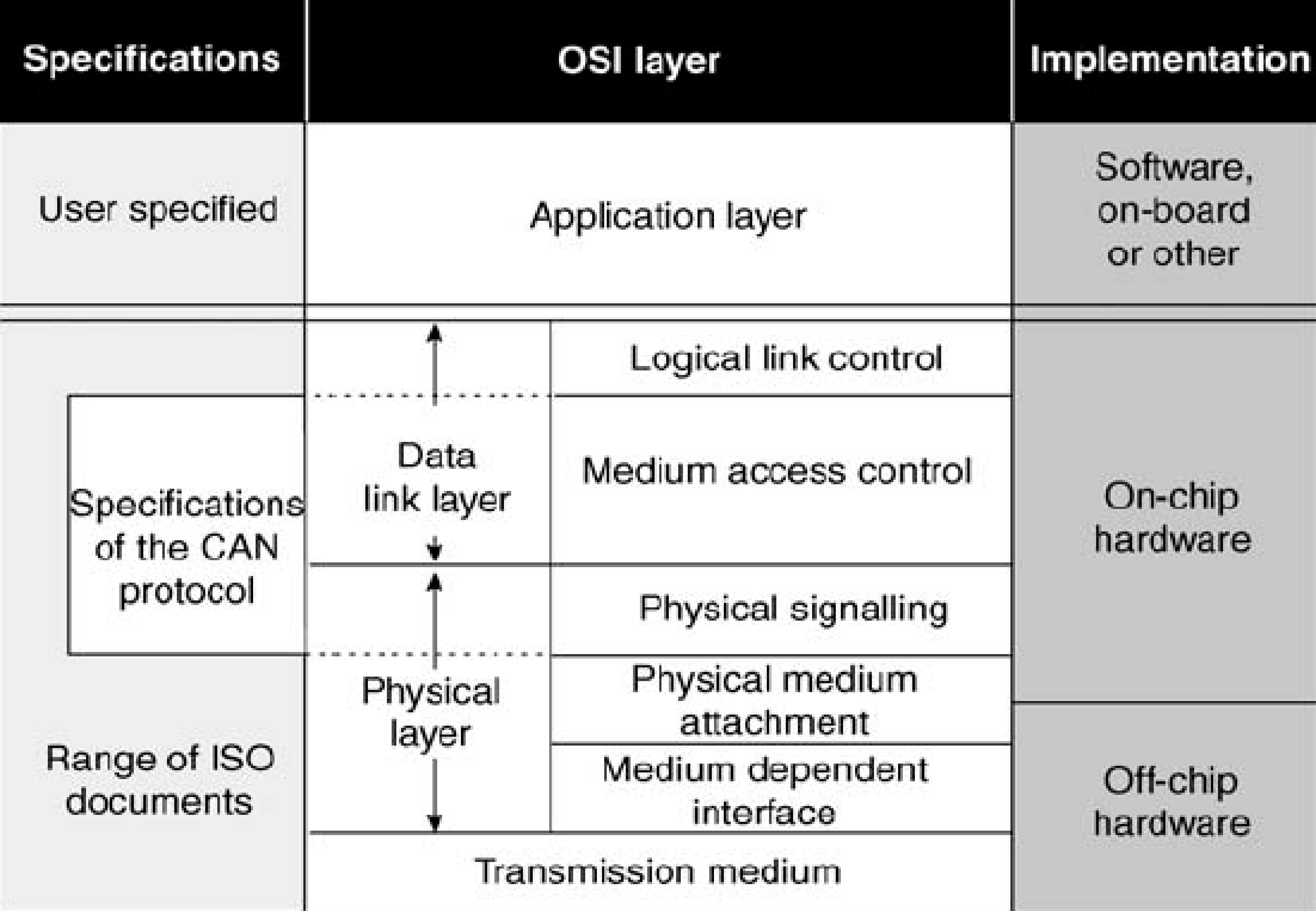
- Serial bus
- Multi-master functionality
- Bitwise arbitration
- Message identification
- Message priority based bus access
- Small messages (Maximum 8 Bytes)
- Maximum data rate: 1 Mbps
- Broadcast communication
- Error detection and retransmission
- Node does not care about configuration of network
- Scalable network

CAN in ISO-OSI Reference Model

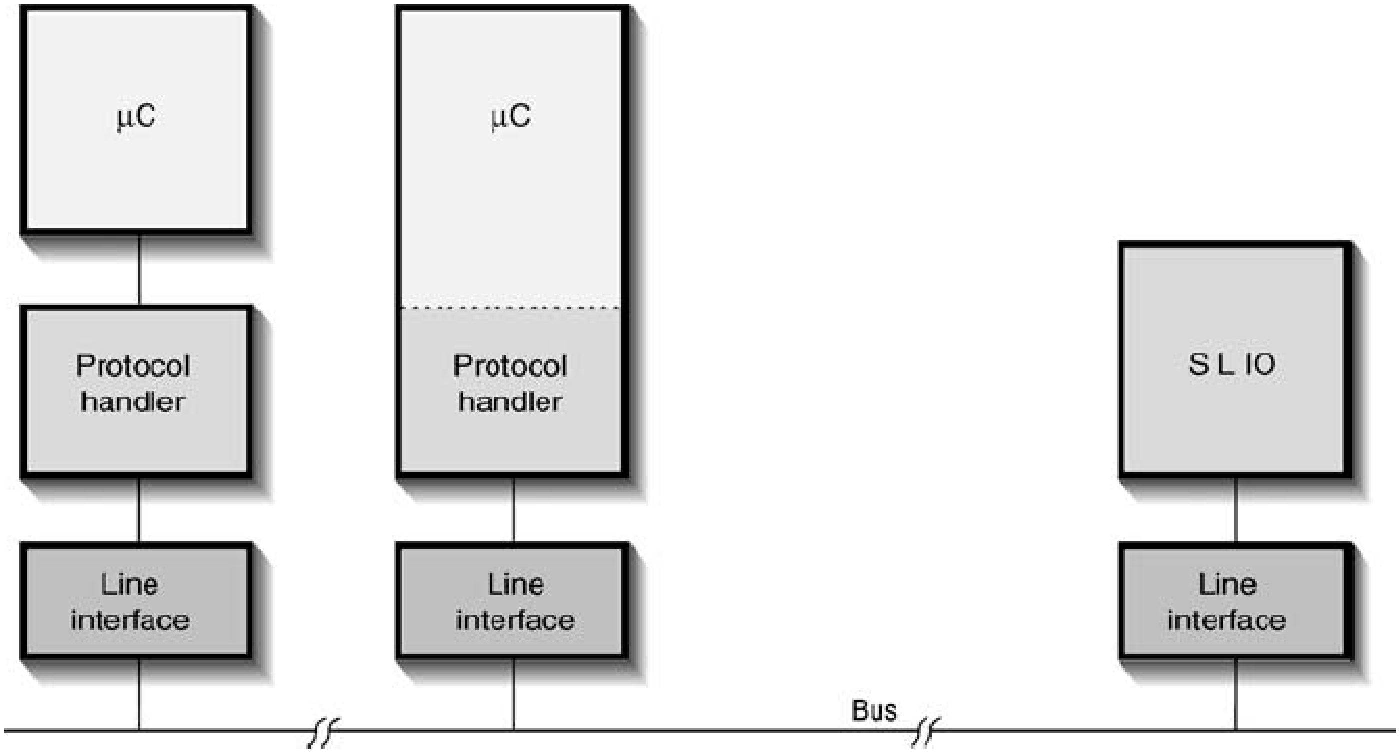



Application Layer
CAN Layers
Data Link Layer
Logical Link Control (LLC) Sublayer
Acceptance Filtering
Overload Notification
Recovery Management
Medium Access Control (MAC) Sublayer
Data Encapsulation/Decapsulation
Frame Coding (Stuffing, Destuffing)
Medium Access Management
Error Detection
Error Signalling
Acknowledgement
Serialization / Deserialization
Physical Layer
Bit Encoding/Decoding
Bit Timing Bus Failure
Synchronization Management
Driver/Receiver Characteristics
CAN Network
CAN Controller
CAN Transceiver


Data link layer
LLC (Logic Link Control)
Acceptance filtering
Overload warning
Recovery Management
MAC (Medium Access Control
Data encapsulation/decapsulation
Frame encoding (stuffing/destuffing
Medium Access Management
Error detection
Error signalling
Acknowledgement
Serialization/deserialization
CAN specification
Fault confinement
Bus malfunction management
Layer 7
Application
Layer 6
Presentation
Layer 5
Session
Layer 4
Transfer
Layer 3
Network
Layer 2
Data link
Layer 1
Physical
Physical layer
PLS (Physical Signalling
Bit coding/decoding
Bit timing
Synchronisation
PMA (Physical Medium Attach.)
Driver/receiver characteristics
MDI (Medium Dependent Interf.)
Connectors


Specifications
OSI layer
Implementation
User specified
Application layer
Software, on-board or other
Logical link control
Data link layer
Medium access control
Specifications of the CAN protocol
On-chip hardware
Physical signalling
Physical layer
Physical medium attachment
Range of ISO documents
Medium dependent interface
Off-chip hardware
Transmission medium

CAN Components



- Stand-alone protocol handlers
- Microcontrollers with on-board CAN handlers
- Line interfaces ('transceivers', or 'drivers')

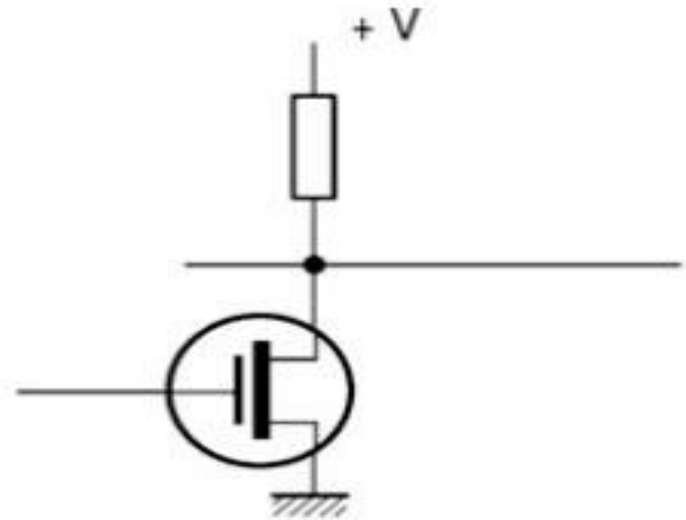
Values of the Bus

- CAN bus can have one of the two complementary logical values defined as
  - ‘0’ as ‘dominant’ and
  - ‘1’ as ‘recessive’
- In case of a simultaneous transmission of a dominant bit and a recessive bit, the resulting value of the bus will be ‘dominant’
- What is the medium mentioned in CAN specification?

Wire links

Recessive = potential recovery

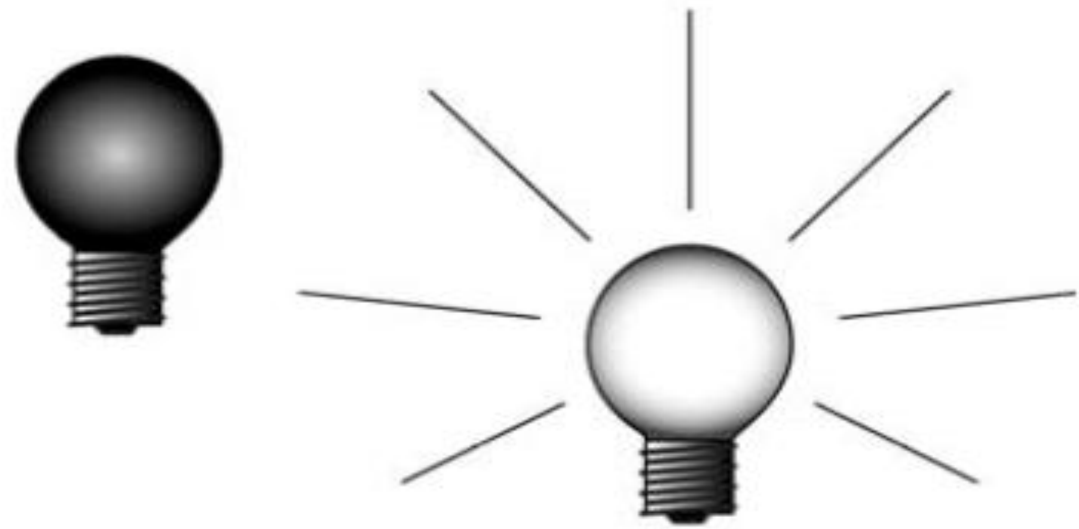
Dominant = conductor grounded



Optical medium

Recessive = light absent

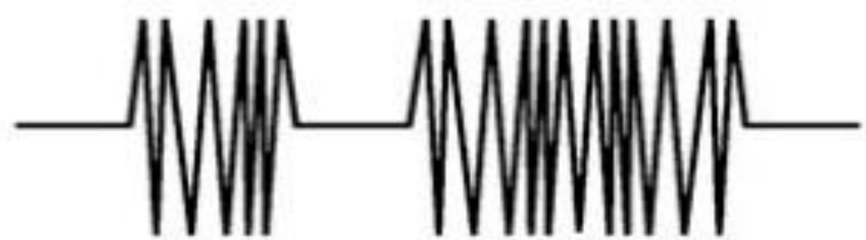
Dominant = light present



Radio medium

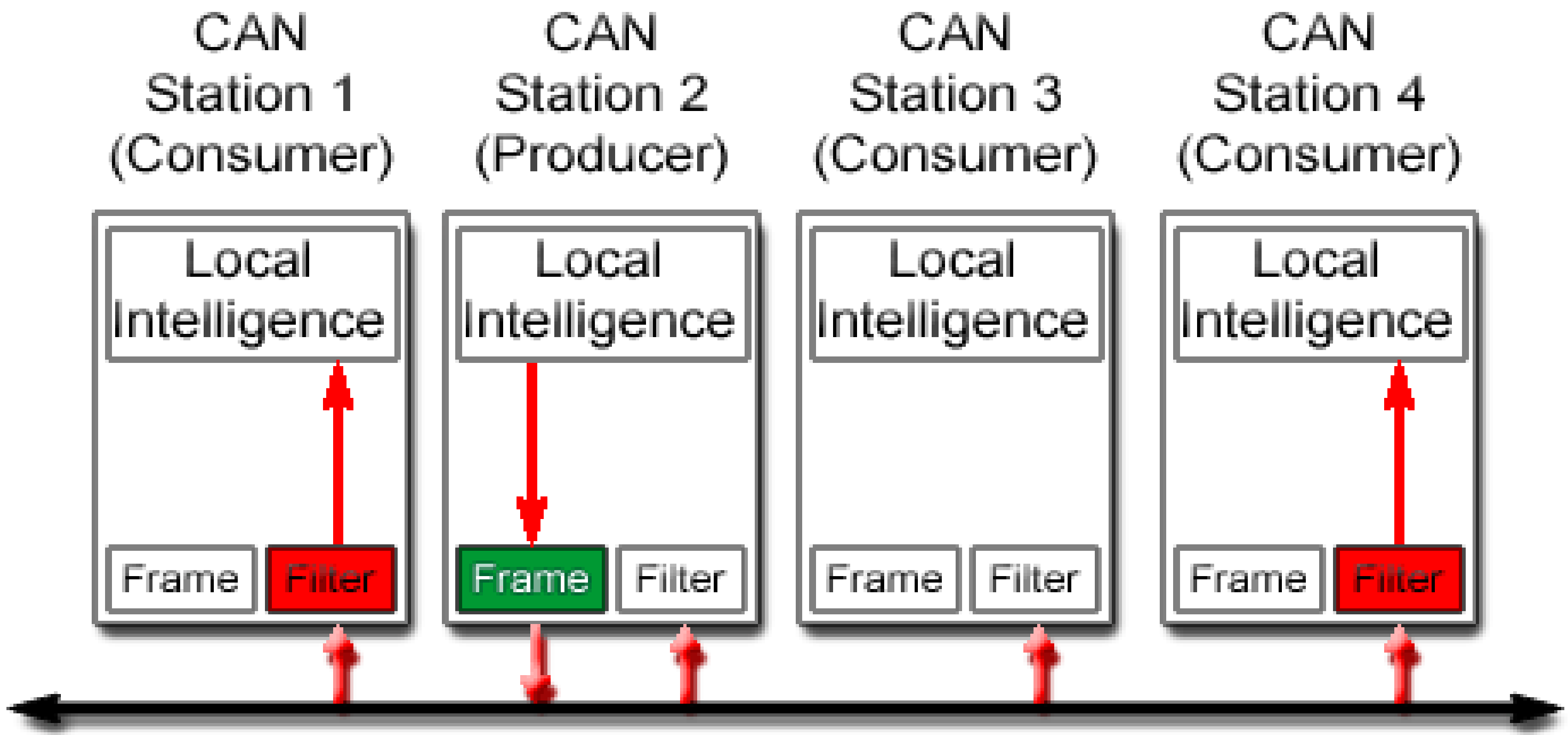
Recessive = carrier absent

Dominant = carrier present

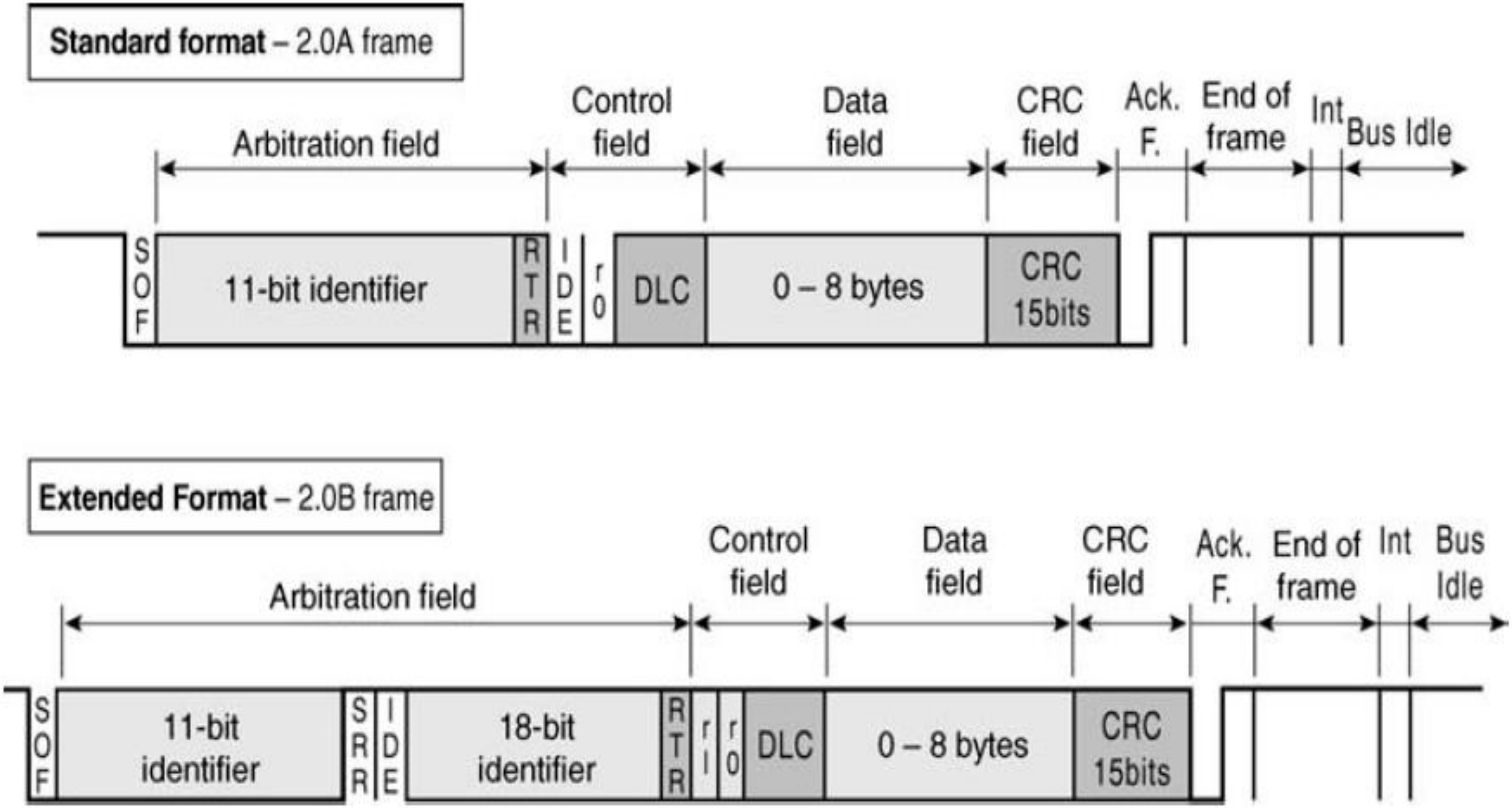


Principles of Data Exchange in CAN

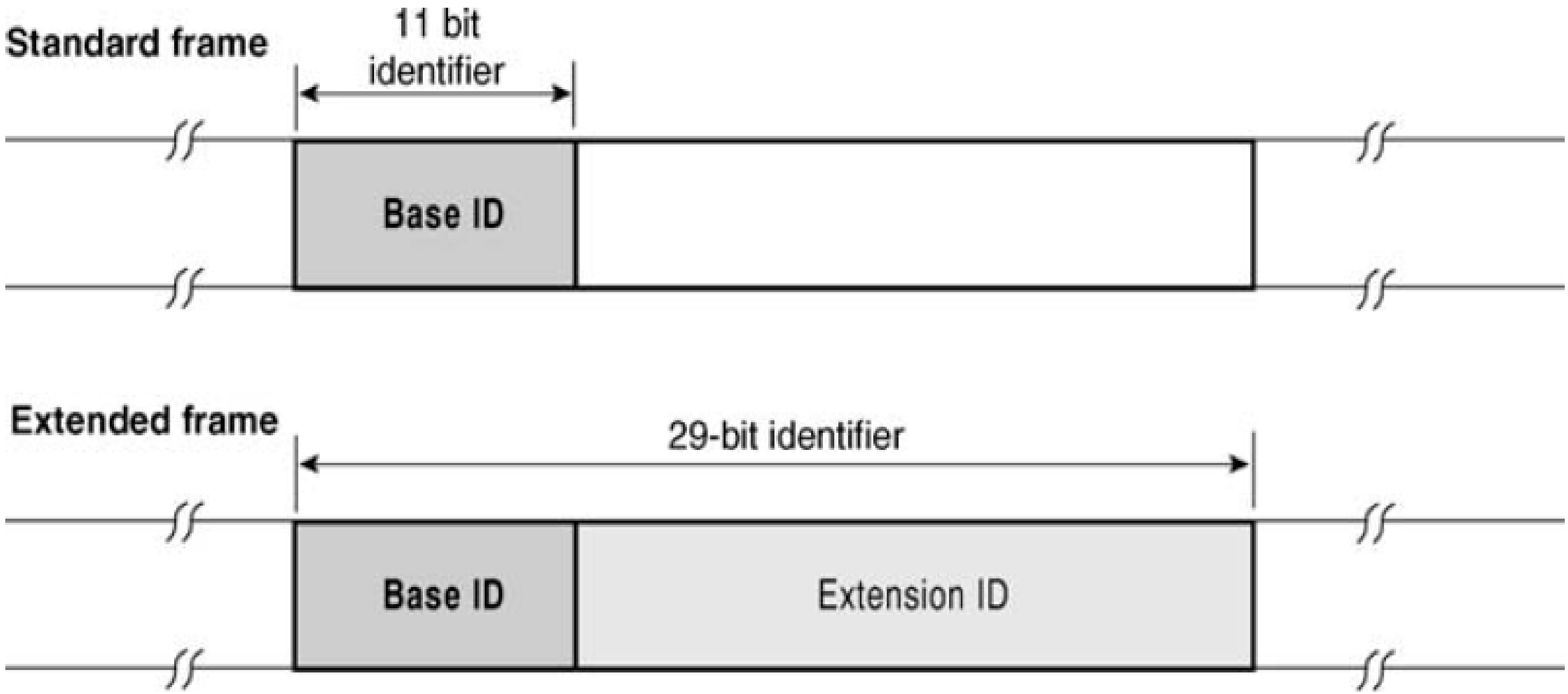
- Broadcast communication mechanism
- Message-oriented transmission protocol



CAN Message Frame Formats

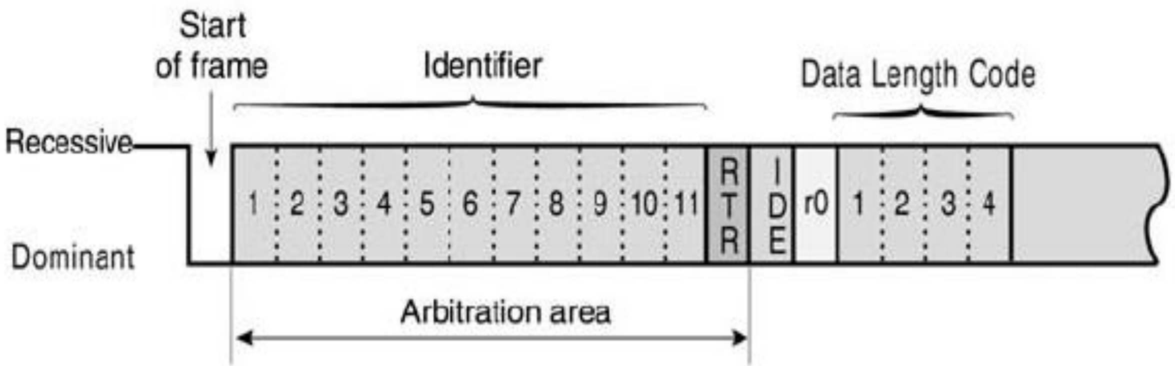


Identifier


Standard frame
11 bit
identifier
Base ID
Extended frame
29-bit identifier
Base ID
Extension ID

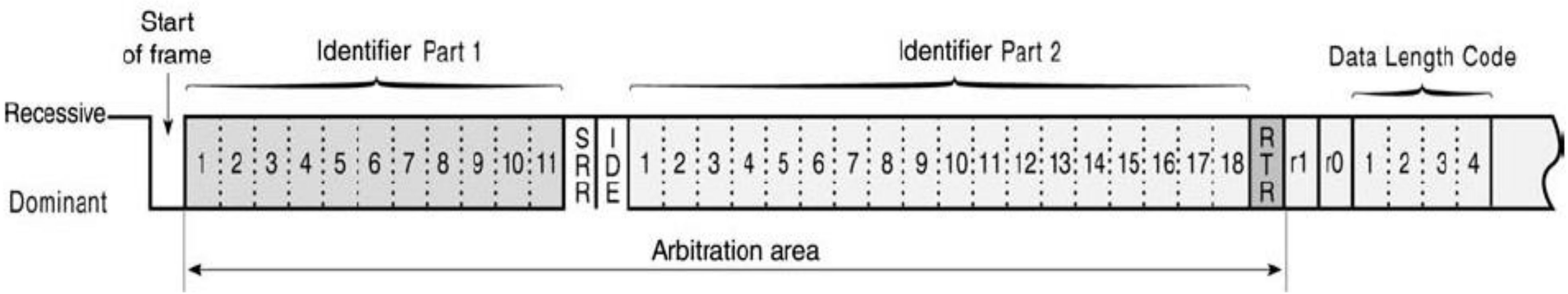
Arbitration

Standard format

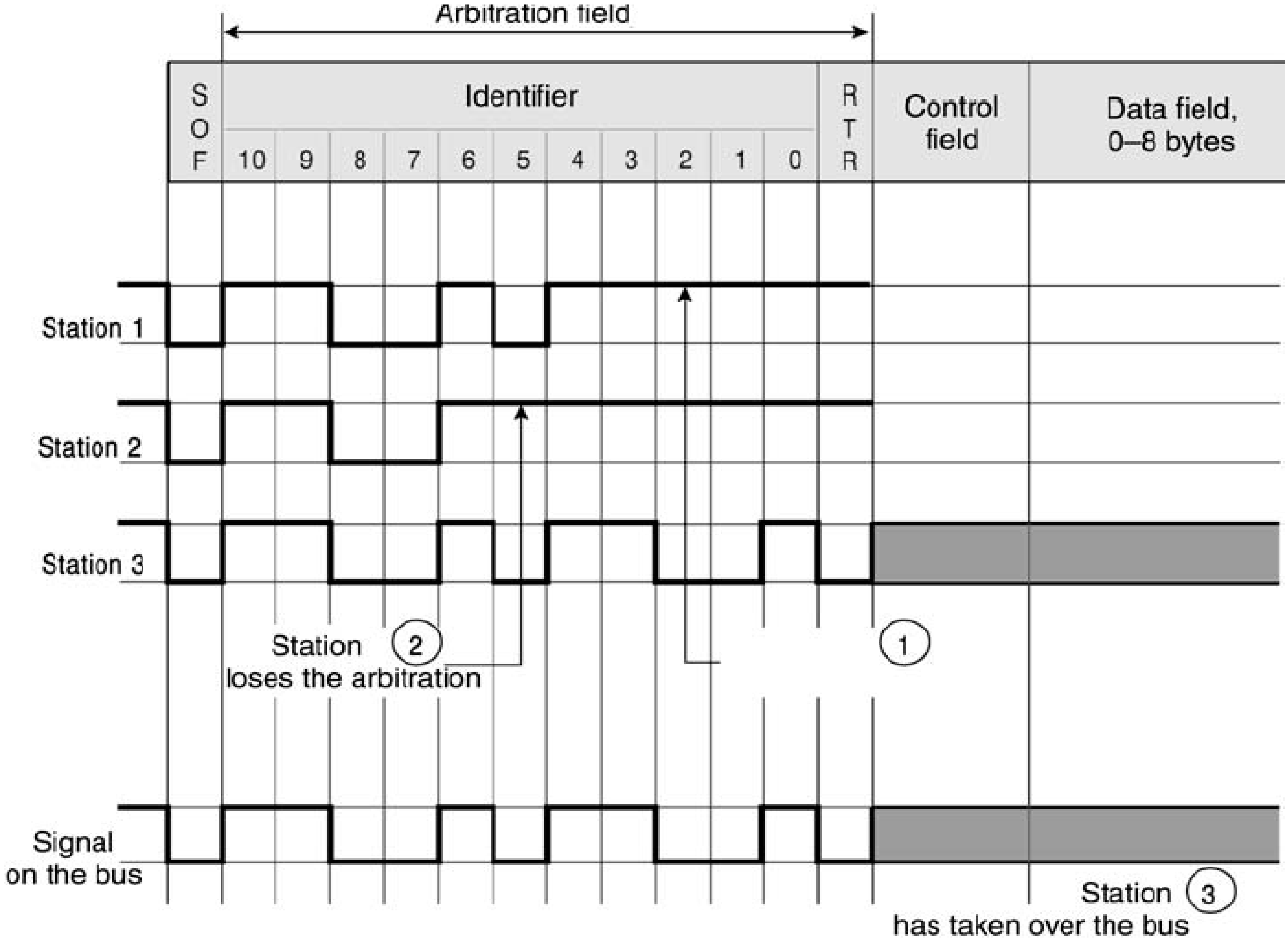


TX	Bus (RX)	Action
Recessive	Recessive	Arbitration procedure
Dominant	Dominant	Arbitration procedure
Recessive	Dominant	Arbitration lost
Dominant	Recessive	Bit error

Extended format



CAN Standard Frame (2.0 A) and CAN Extended Frame (2.0 B)


Arbitration field
SOF
Identifier
10
9
8
7
6
5
4
3
2
1
0
RTR
Control field
Data field, 0–8 bytes
Station 1
Station 2
Station 3
Station 2 loses the arbitration
1
Signal on the bus
Station 3 has taken over the bus

Compatibility between CAN 2.0A and CAN 2.0B

- CAN 2.0B controllers are completely backward compatible with CAN 2.0A

- There are two types of CAN 2.0 A controllers
  - Only CAN 2.0A (flag error on reception of CAN 2.0B frames)
  - CAN 2.0A active, CAN 2.0B passive (acknowledge receipt of CAN 2.0B messages and then ignore them)

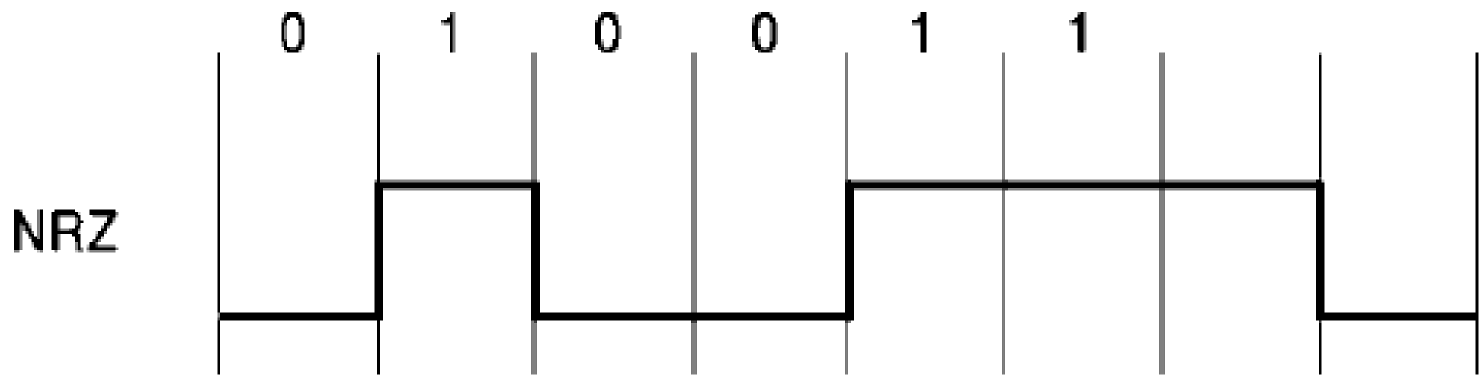
Number of Available Addresses

- With 11 bit identifier of CAN standard format, number of available addresses is 2^11 = 2048

- With 29 bit identifier of CAN extended frame format number of available addresses is more than 536 million

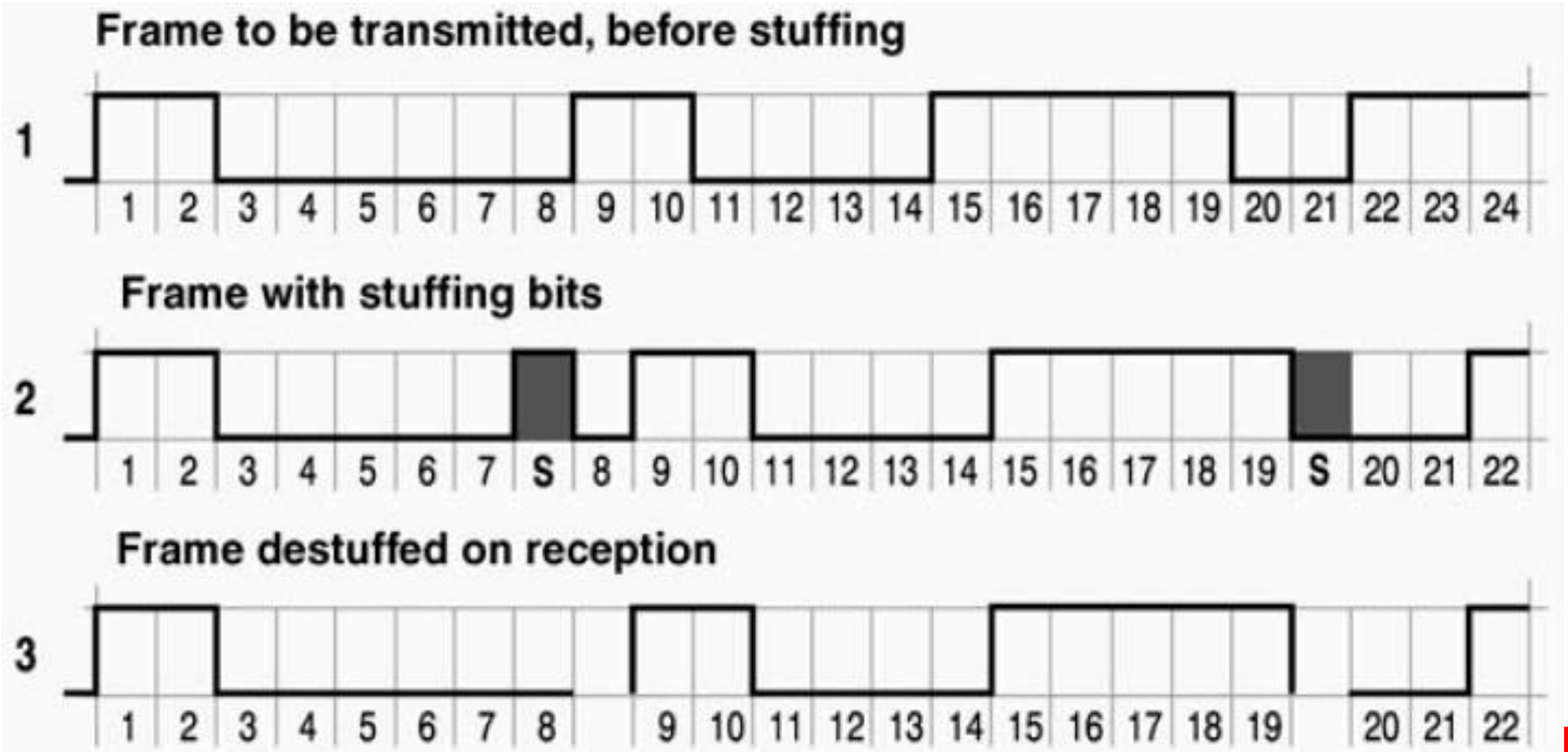
Coding

- Non Returned to Zero (NRZ) coding method is adopted

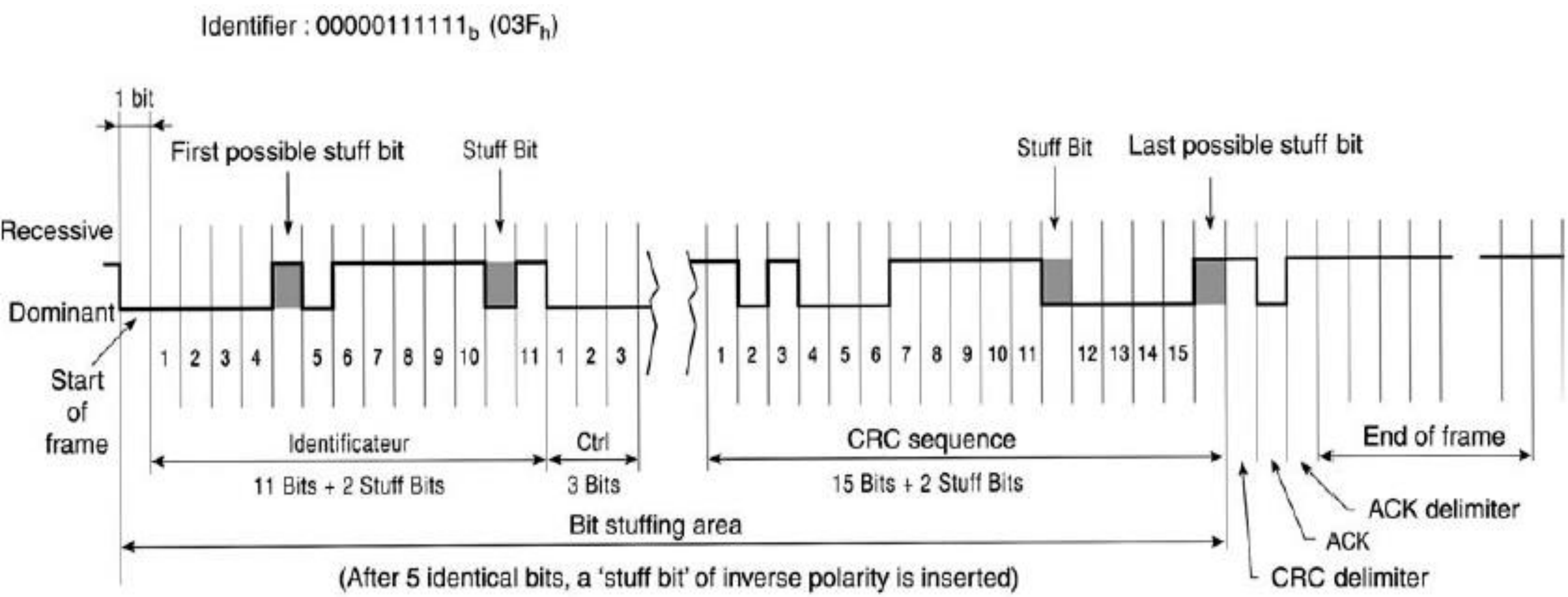


Bit Stuffing

- After 5 bits of identical value (either dominant or recessive), a supplementary bit having an intentionally opposite value is introduced

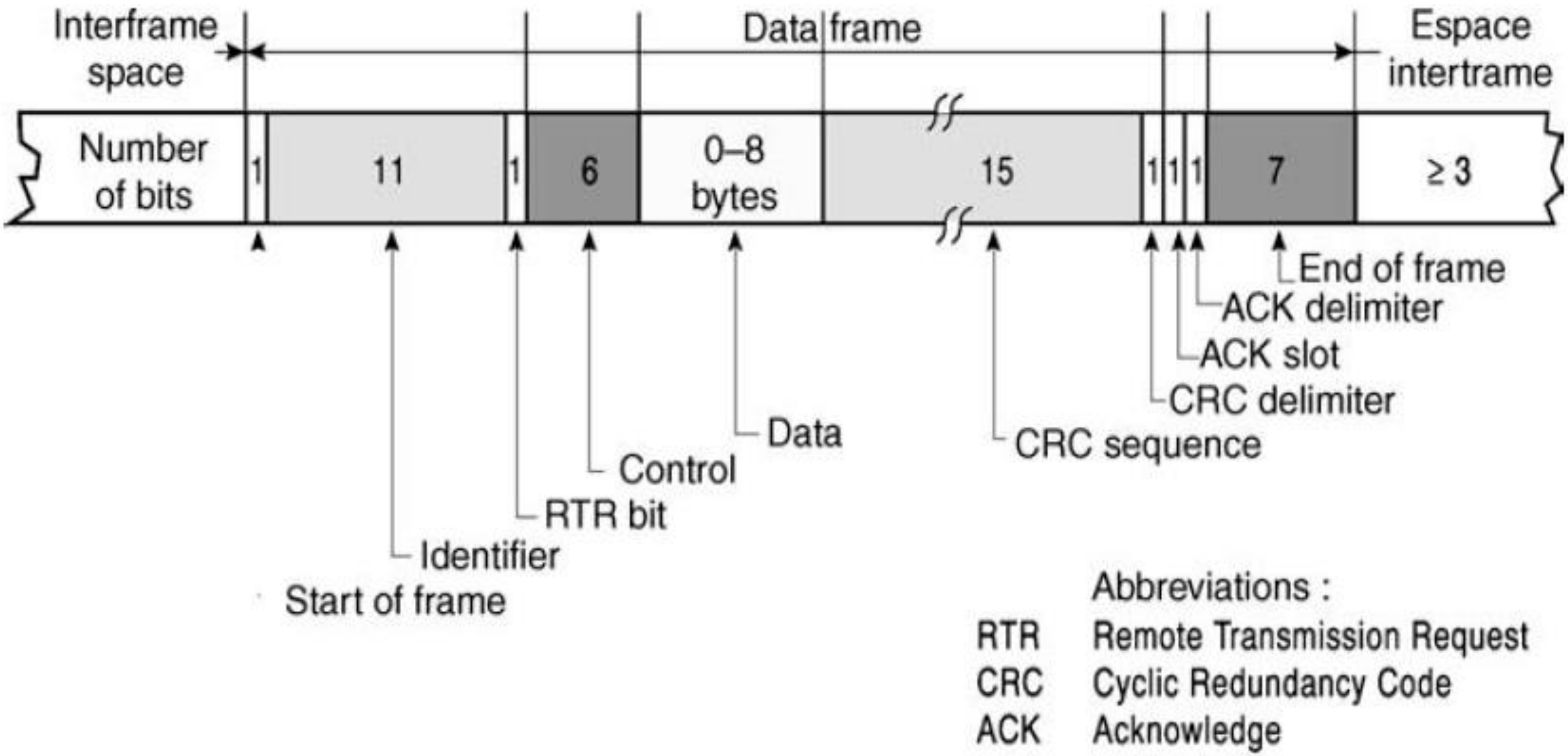


Example of Bit Stuffing in CAN Frame

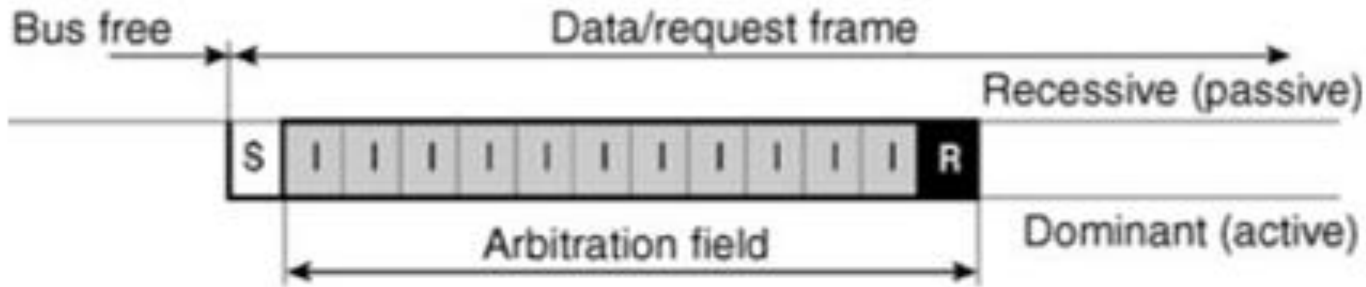


CAN « Bit Stuffing »

Components of CAN Frame



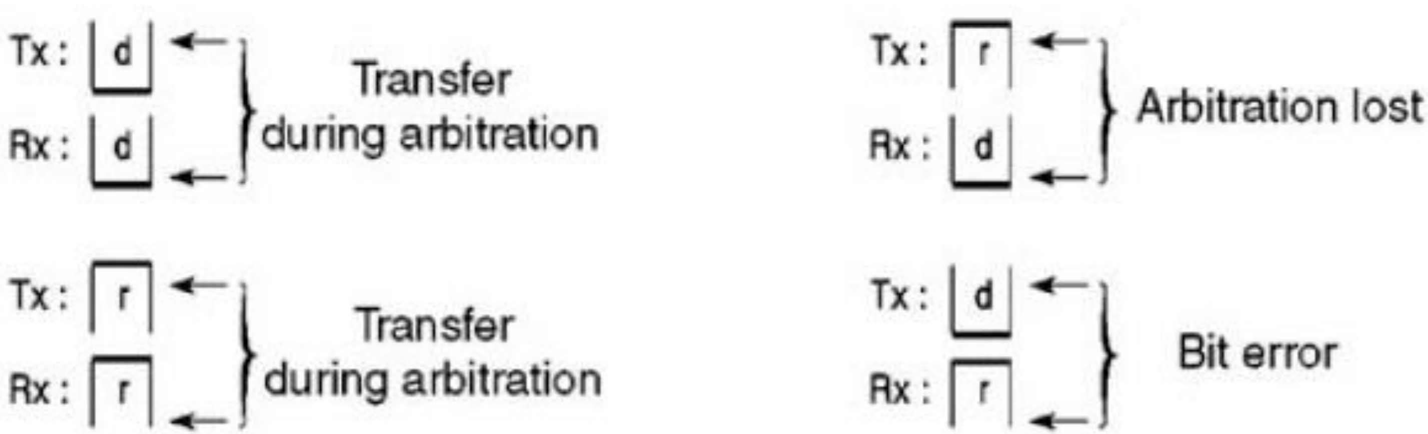
Arbitration Field



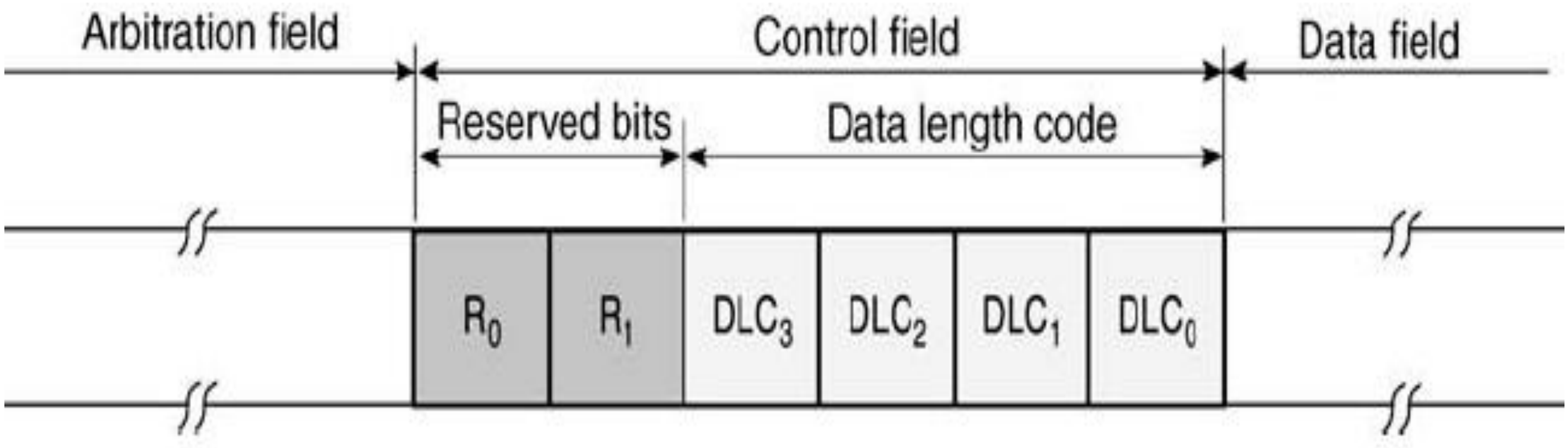
I: 1 of 11 identifier bits

R: RTR bit
- recessive: request frame
- dominant: data frame

S: frame start bit

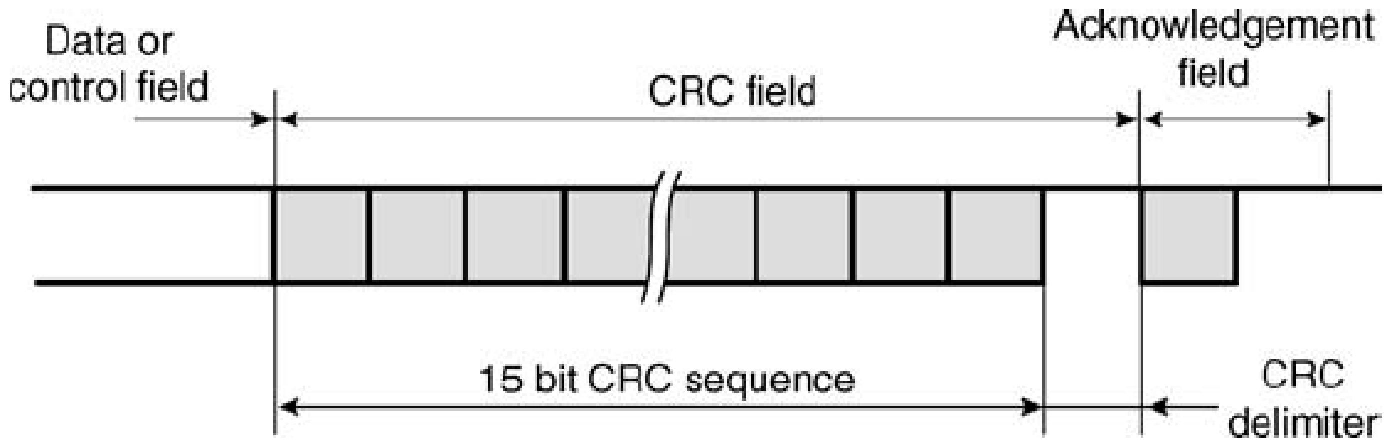


Control Field



- Data Length Code bits determine how many data bytes are included in the message

CRC Field

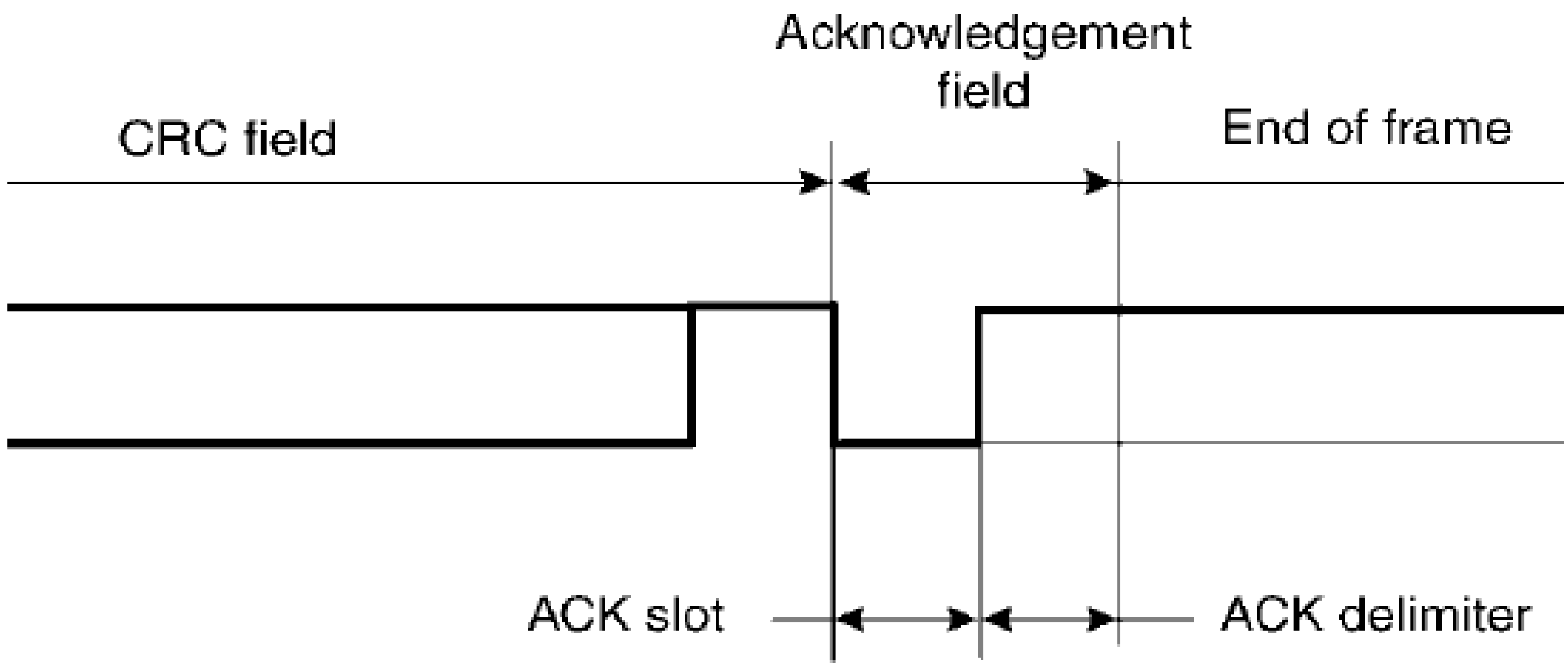


$$g(X) = X^{15} + X^{14} + X^{10} + X^{8} + X^{7} + X^{4} + X^{3} + 1$$

$$g(X) = 1100010110011001$$

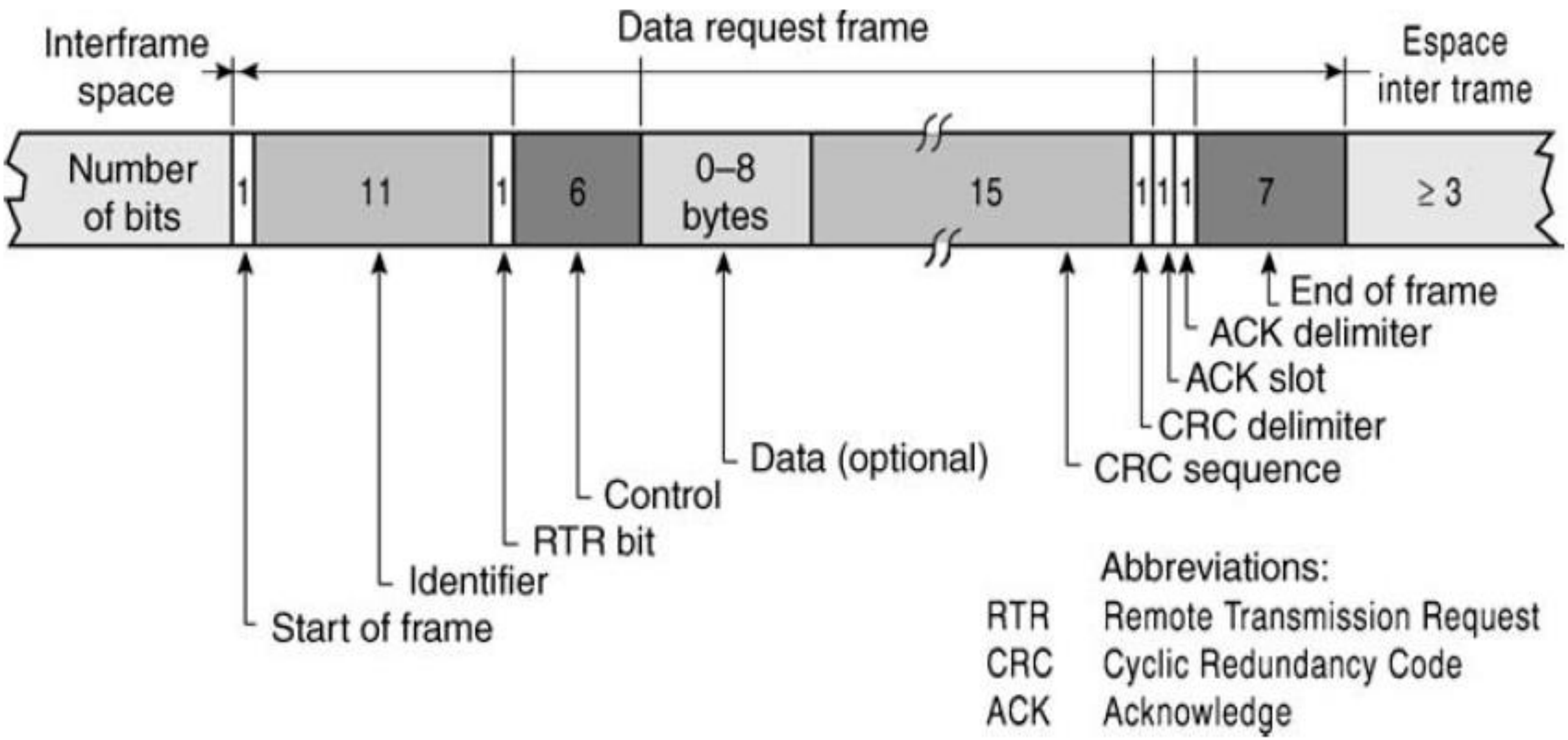
- The CRC Field consists of a 15-bit CRC field and a CRC delimiter, and is used by receiving nodes to determine if transmission errors have occurred

Acknowledgement Field



- Acknowledgement field indicates if the message was received correctly

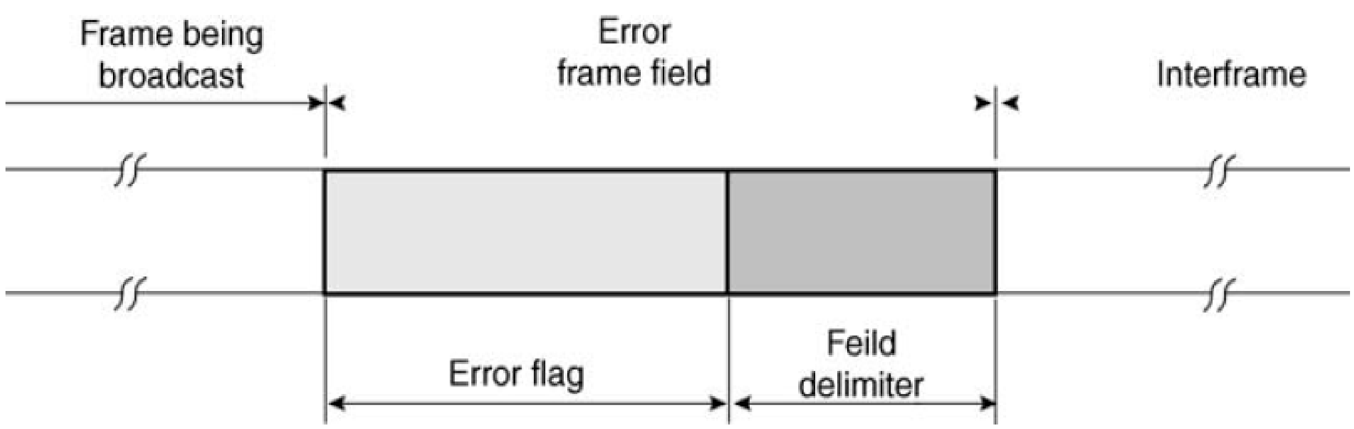
Remote Transmit Request (RTR)



What if data/request frames with same identifier are on the bus simultaneously?

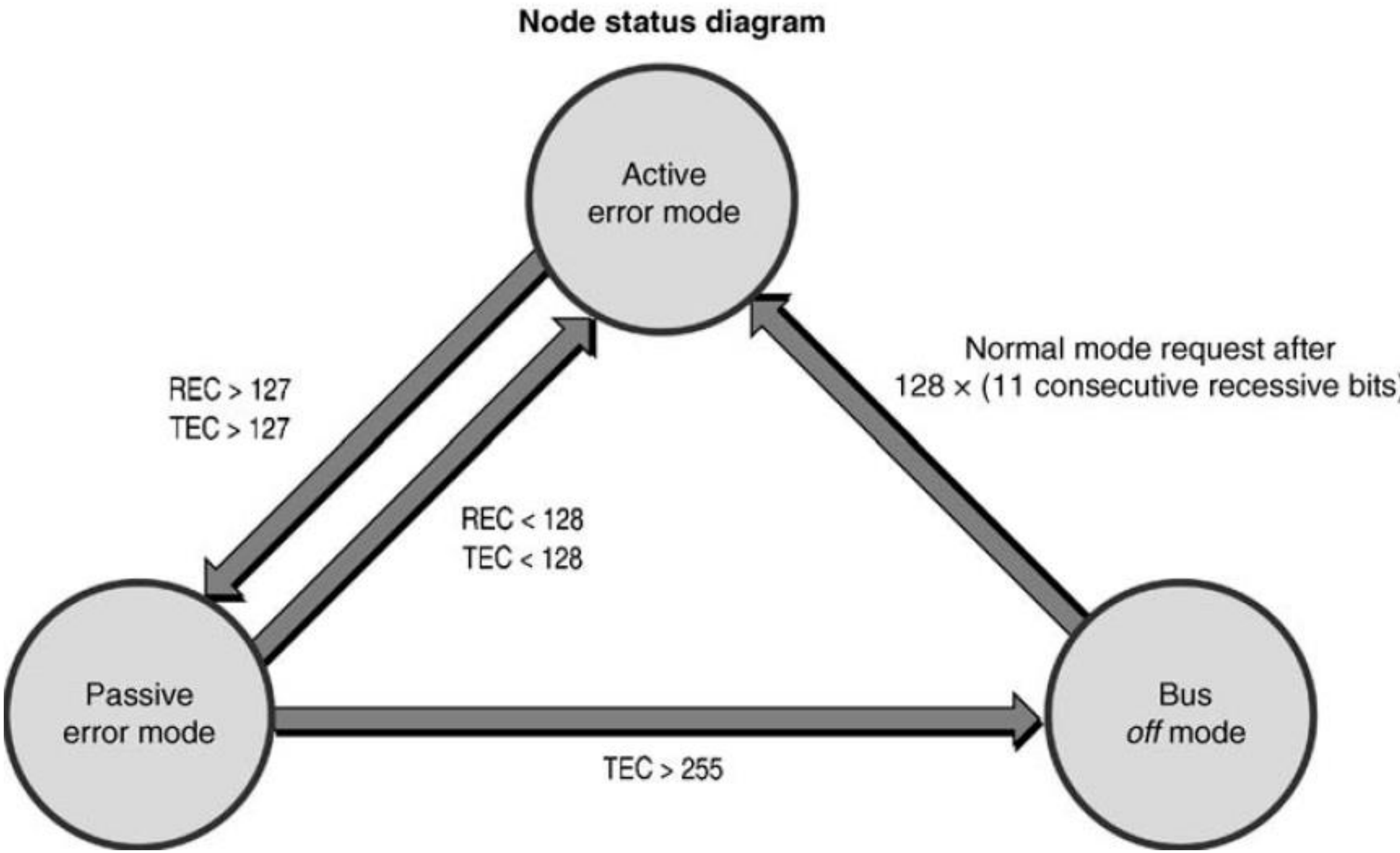
Detecting and Signaling Errors

- Cyclic Redundancy Check (CRC)
- Format Error
- Acknowledgement Errors
- Bit Error



Error Frame

Error Processing



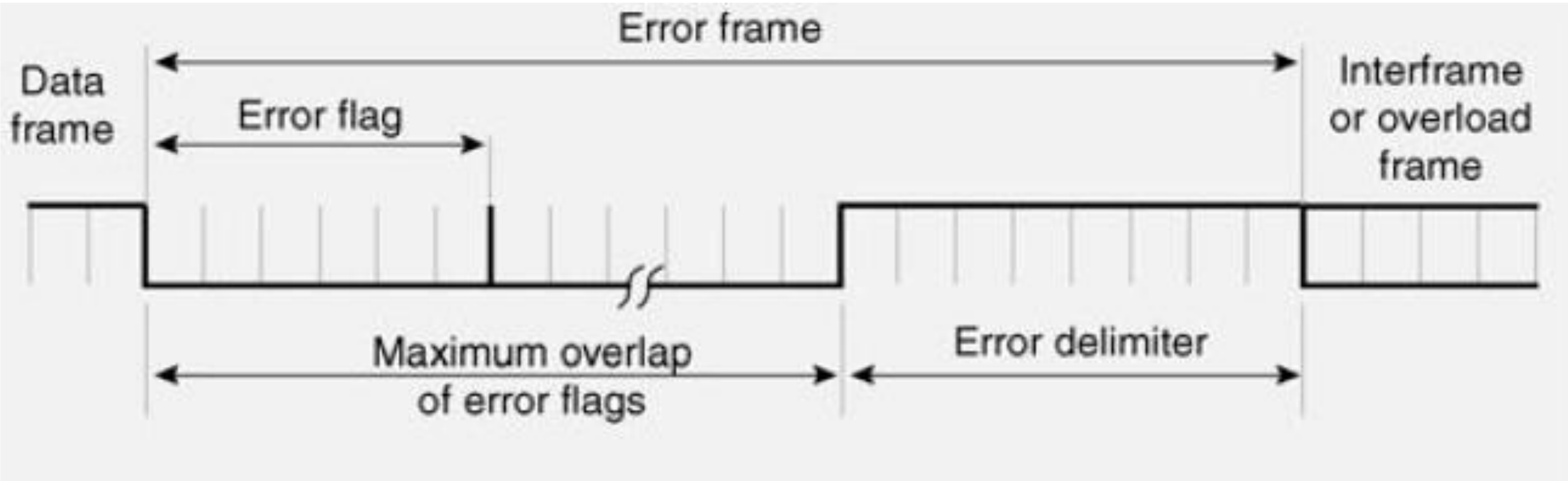
Warning limit 96

Error Frames

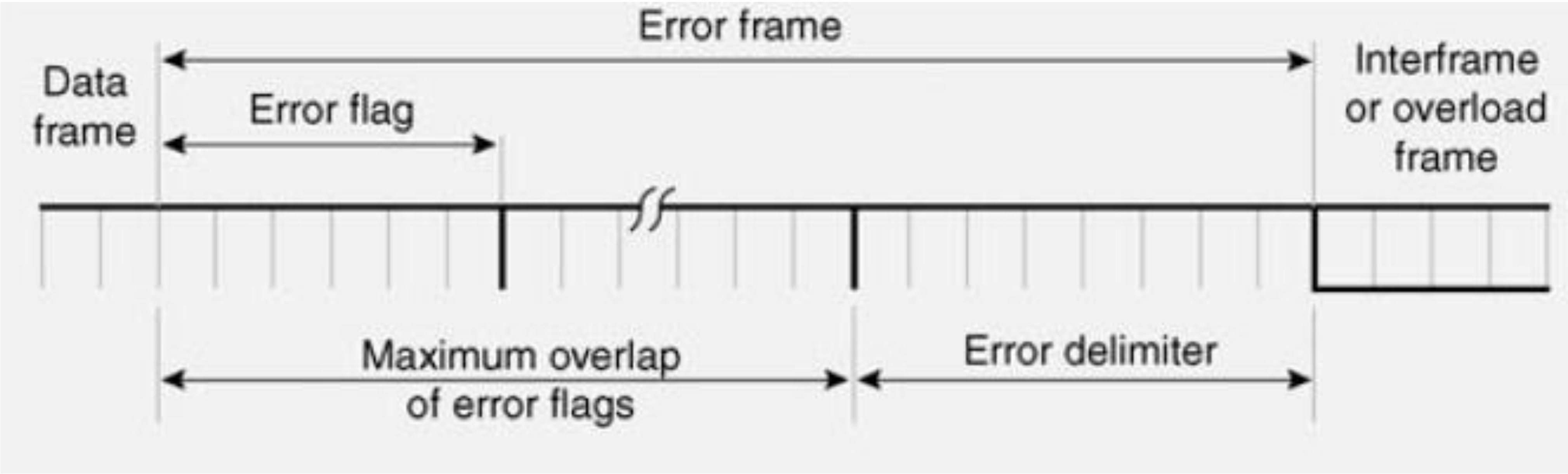
- Active error frame
  - Error-Active node continues to tx and rx normally
  - Transmitted when the node is in active error state
  - Frame comprises of six consecutive dominant bits

- Passive error frame
  - Passive error nodes continue to tx and rx normally
  - These nodes transmit passive error frame comprising of six recessive bits

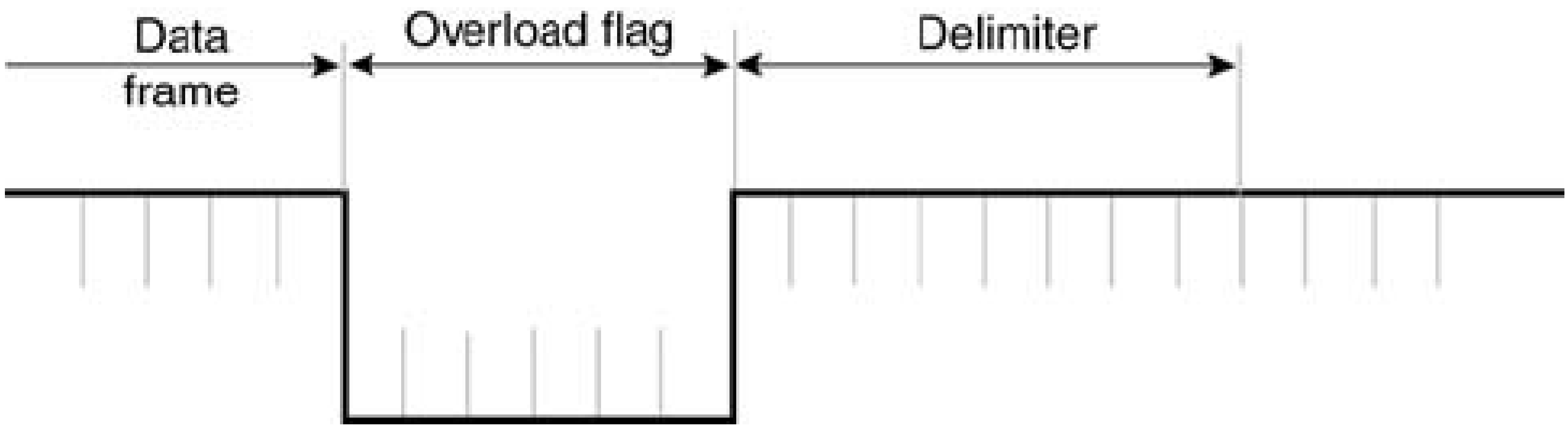
Active Error Frame



Passive Error Frame



Overload Frame



- Can appear at the EOF or end of error delimiter
- During intermission period

CAN Sleep and Wake Up

- Nodes can transit between sleep and wake up modes for power saving

- After coming to wake up mode, node first needs to synchronize by observing 11 consecutive recessive bits

CAN with Flexible Data-Rate

Specification

Version 1.0

(released April 17th, 2012)

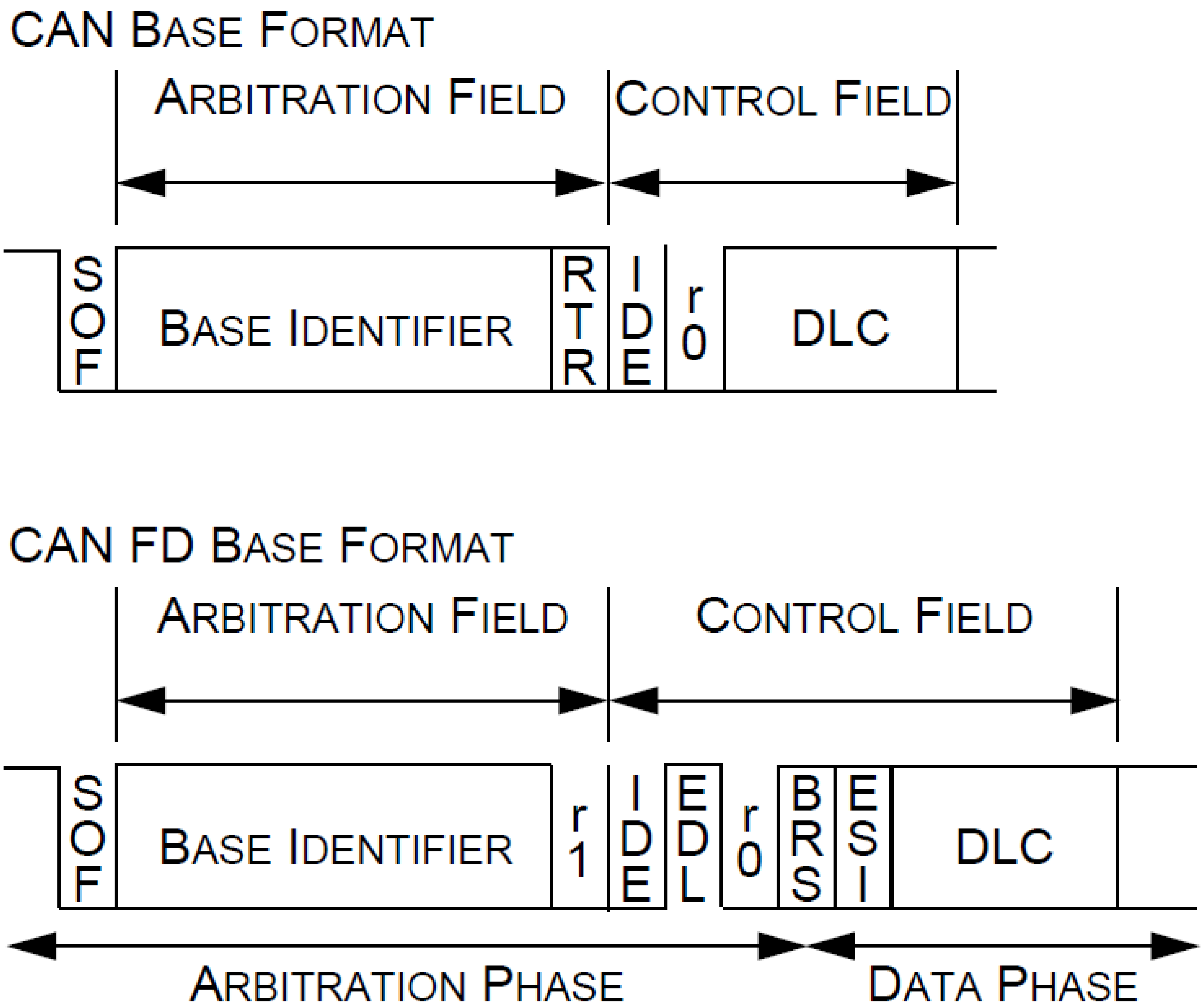
Characteristics of CAN FD

- Allows data rates up to 15 Mbps
- Allows payloads of up to 64 bytes per frame
- CAN FD shares the physical layer, with the CAN protocol as defined in the BOSCH CAN Specification 2.0
- Two new control bits in the CAN FD frame:
  - First enabling the new frame format with different data length coding
  - Second optionally switching to a faster bit rate after the arbitration
- Longer CRC polynomials

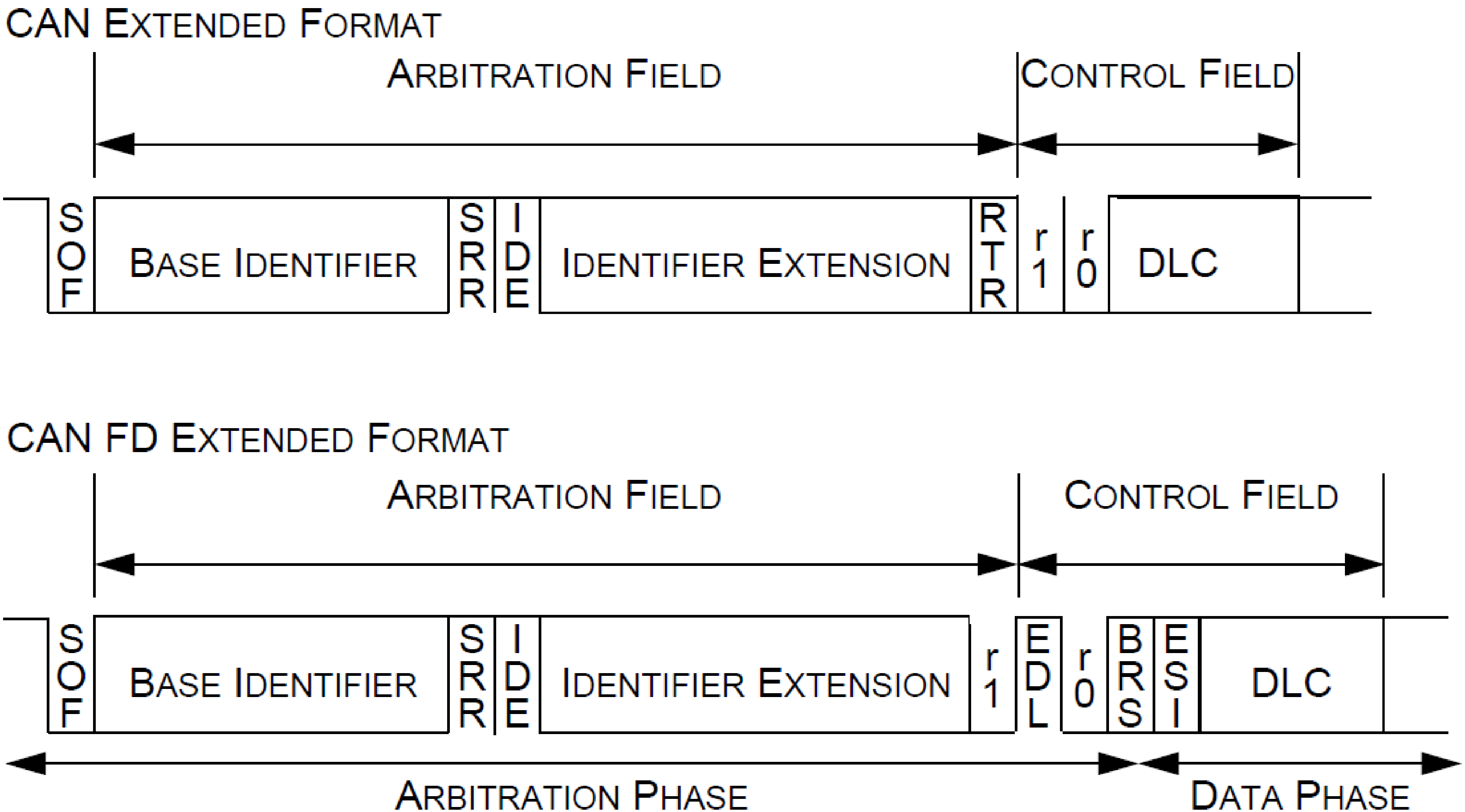
Characteristics of CAN FD

- Backward compatible with 2.0A and 2.0B
- There is no REMOTE FRAME in the CAN FD format
- Each CAN FD node however is able to transmit a REMOTE FRAME in the standard CAN format
- CAN FD Frame formats
  - CAN FD BASE FORMAT: 11 bit long identifier and dual bit rate
  - CAN FD EXTENDED FORMAT: 29 bit long identifier and dual bit rate

CAN FD Base Frame Format



CAN FD Extended Frame Format



Control Field

- RTR bit is replaced with the *dominant* reserved bit r1
- No remote frames in CAN FD format
- In CAN base format the control field consists of the bits
  - IDE -0
  - r0- 0, reserved for future expansion
  - 4 bits wide data length code (DLC)
- In CAN FD base format the control field consists of the bits
  - IDE
  - EDL- Extended Data Length, recessive, distinguishes b/w CAN and CAN FD frames
  - r0 - 0, reserved for future expansion
  - BRS – Bit Rate Switch, if recessive then bit rate is switched from standard bit rate to preconfigured alternate bit rate, if dominant then bit rate is not switched
  - ESI- Error state indicator, transmitted dominant by error active nodes, recessive by error passive nodes
  - 4 bits wide data length code (DLC)

CRC in CAN FD

$$(x^{15}+x^{14}+x^{10}+x^{8}+x^{7}+x^{4}+x^{3}+1)$$

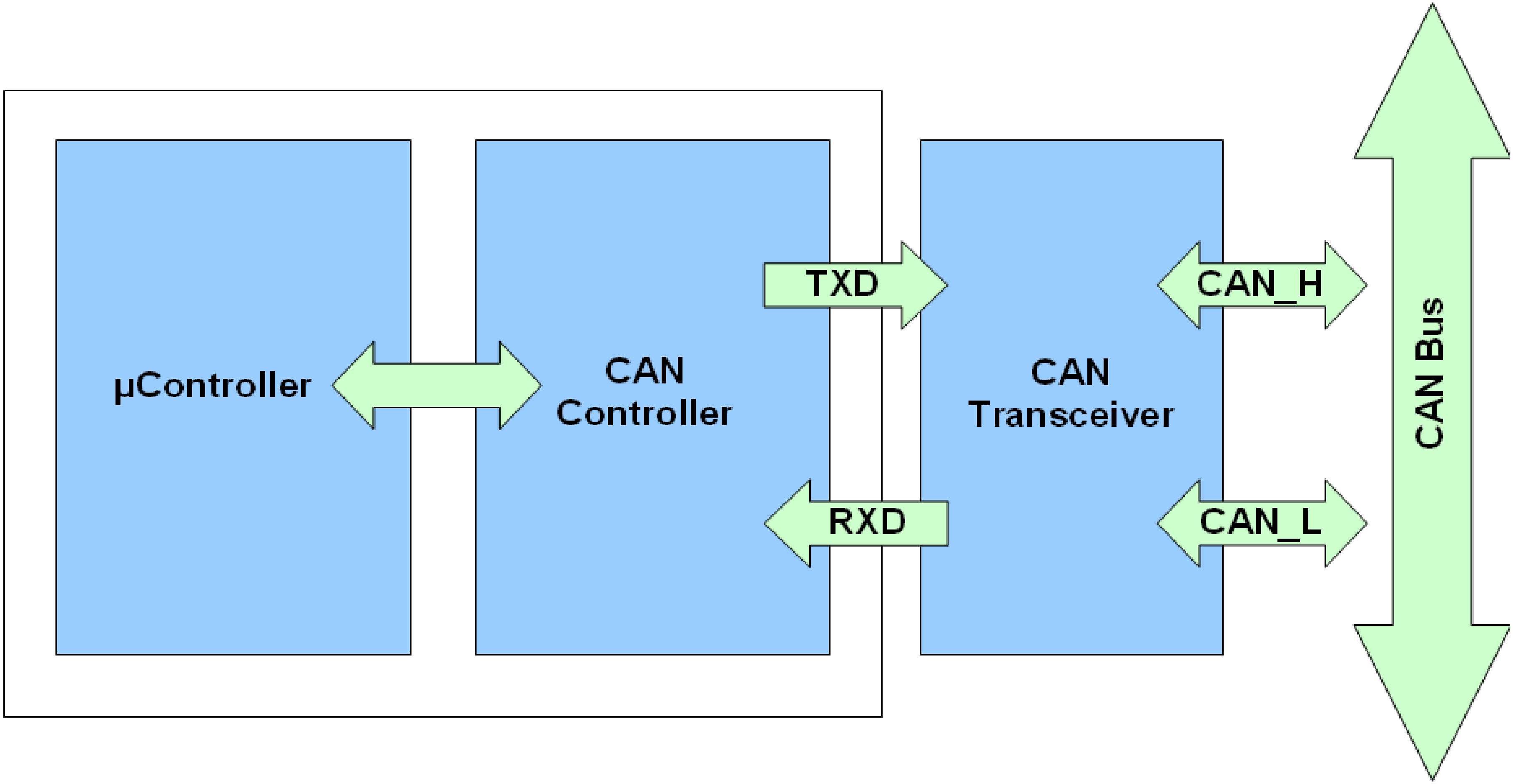
$$(x^{17}+x^{16}+x^{14}+x^{13}+x^{11}+x^{6}+x^{4}+x^{3}+x^{1}+1)$$

$$(x^{21}+x^{20}+x^{13}+x^{11}+x^{7}+x^{4}+x^{3}+1)$$

- Based on values of EDL bit and DLC, appropriate CRC polynomial is considered by transmitter and receivers

CAN Physical Layer

Typical CAN Node



CAN Bus is a simple 2-wire differential serial bus
CAN Bus is terminated on each side by a 120 Ohm resistor

Components of CAN-PHY

- The OSI layer model defined for CAN specification consists of the following physical layers:

- Physical layer (layer 1), PL:
  - Physical signaling sublayer, PLS (bit coding, bit timing, synchronization)
  - Physical medium attachment sublayer, PMA (characteristics of the command stages (drivers) and reception stages)
  - Medium dependent interface sublayer, MDI (connectors)

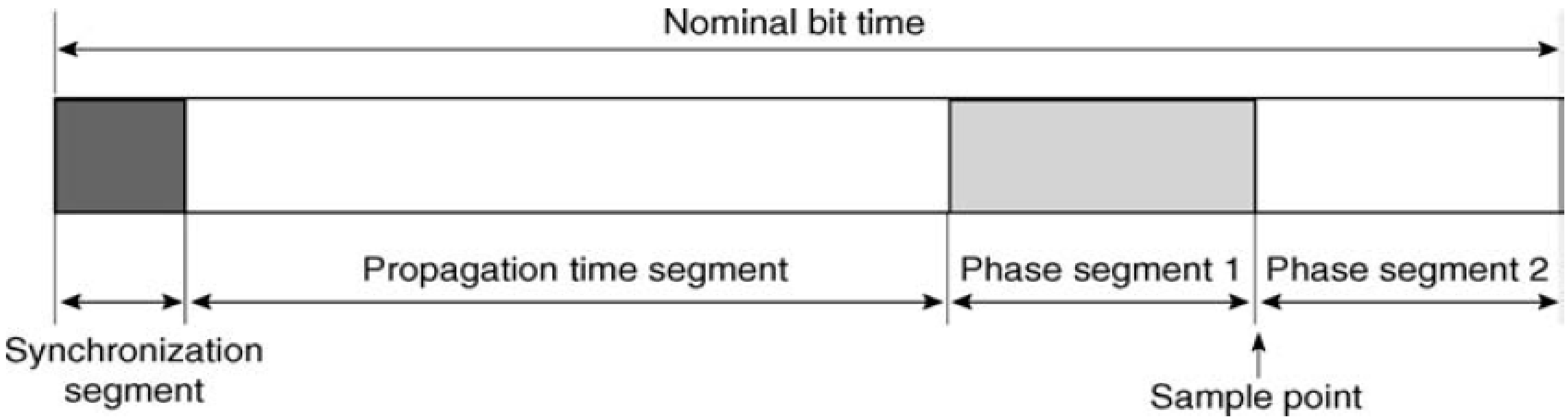
- Transmission medium:
  - bus cable
  - bus termination networks

Nominal Bit-Time

- Normally constructed electronically from a specified number of 'system' cycle pulses (based on internal clock of each station)

- By definition, it is equal to

$$\text{Nominal bit time} = \frac{1}{\text{Nominal bit rate}}$$



Nominal Bit-Time

- Synchronization segment
  - part of the bit time used to synchronize the various nodes present on the bus
  - leading edge of an incoming bit should appear within this time segment
- Propagation time segment
  - Designed to allow for and compensate delay times due to propagation of the signal over the medium used
- Phase buffer segments 1 and 2
  - Compensate for the variations, errors or changes in the phase and/or position of the edges of the signals while travelling over the medium
  - Can be either shortened or extended during resynchronization

Bit-Timing and Synchronization

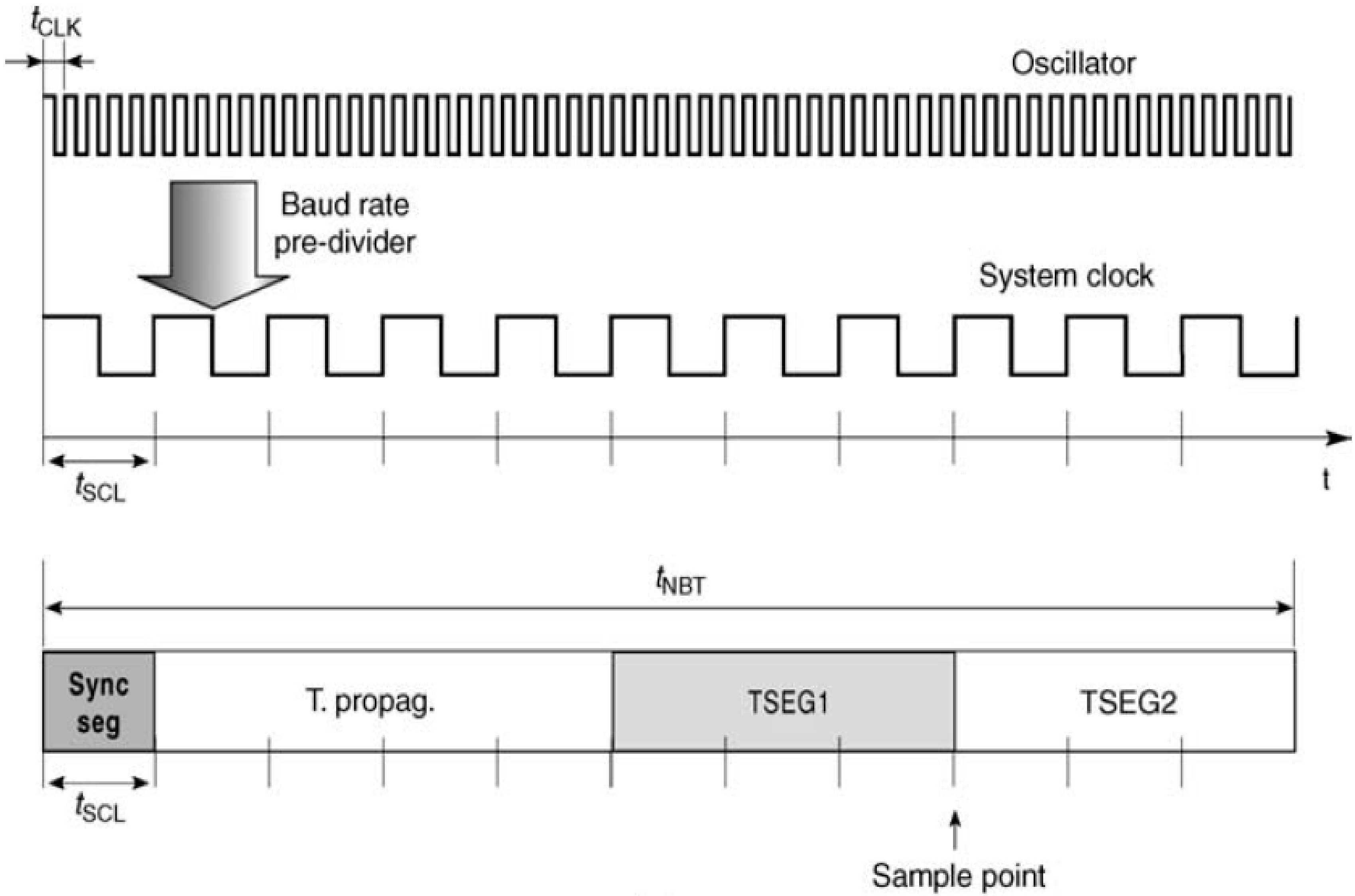
- Two types of synchronization
  - Hard synchronization
  - Resynchornization

- Minimum time quantum of the station
  - derived from the station's clock (oscillator)

$$\text{Time quantum} = m \times \text{Minimum time quantum}$$

- Time quantum of the bit time
  - a fixed unit of time
  - derived from the minimum time quantum of the CAN bit

Bit-Timing and Synchronization



Bit-Timing and Synchronization

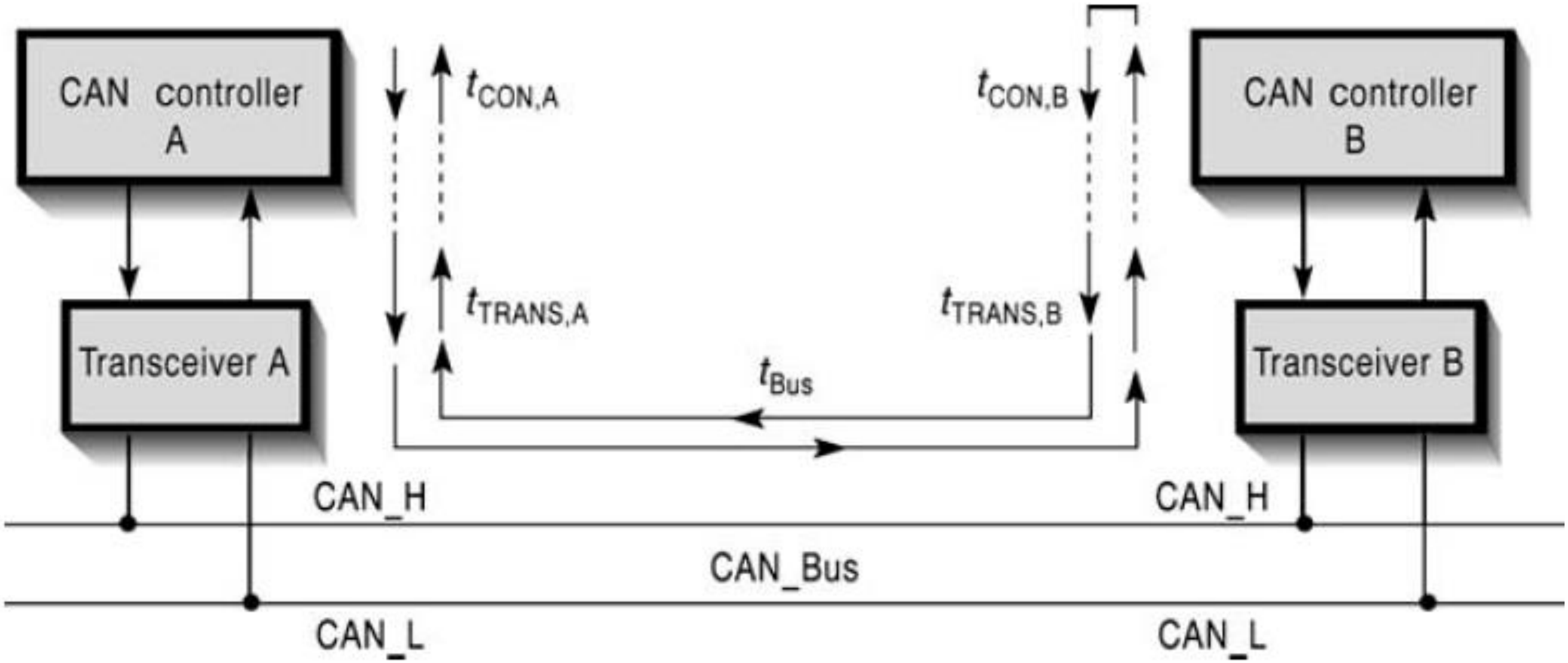
- Synchronization time: 1 time quantum (fixed value)
- Propagation time segment: 1 to 8 time quanta
- Segment 1 of the phase buffer: 1 to 8 time quanta
- Information processing time: ≤ 2 time quanta
- Segment 2 of the phase buffer: Value of segment 1 + information processing time
- For best adaptation to CAN network, these values are generally programmable through special registers

ज्ञानं विज्ञानं च भक्तिसहितं

Factors Affecting Data Transmission in the Medium

- Propagation speed of signal in chosen medium
- Length of the medium
- Internal delay times of transmission and reception circuit components
- Choice of medium (electromagnetic compatibility; EMI/EMC)

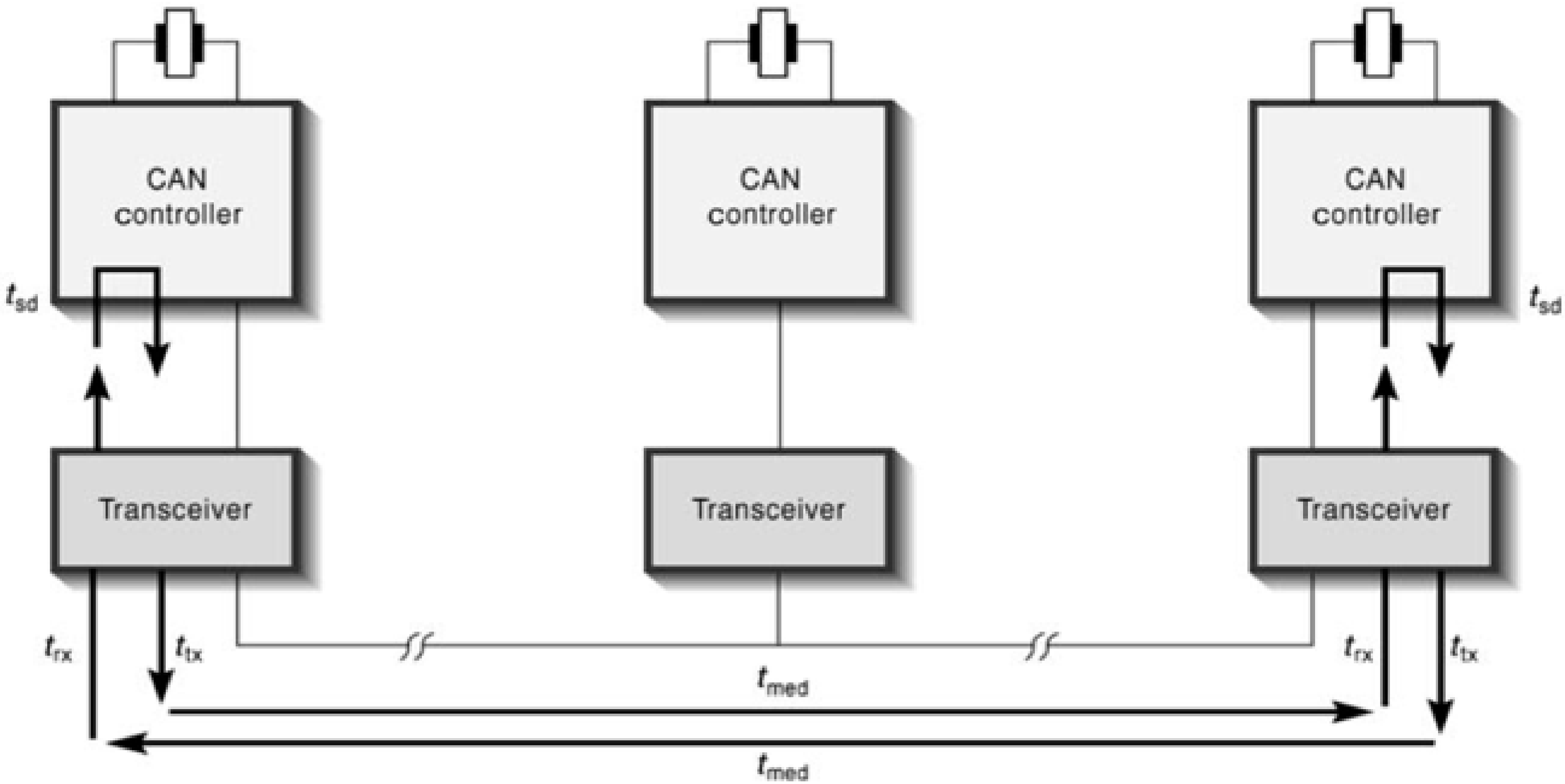
Propagation Time



In a single network, the longest distance between two CAN controllers determines the maximum 'outgoing' propagation delay time

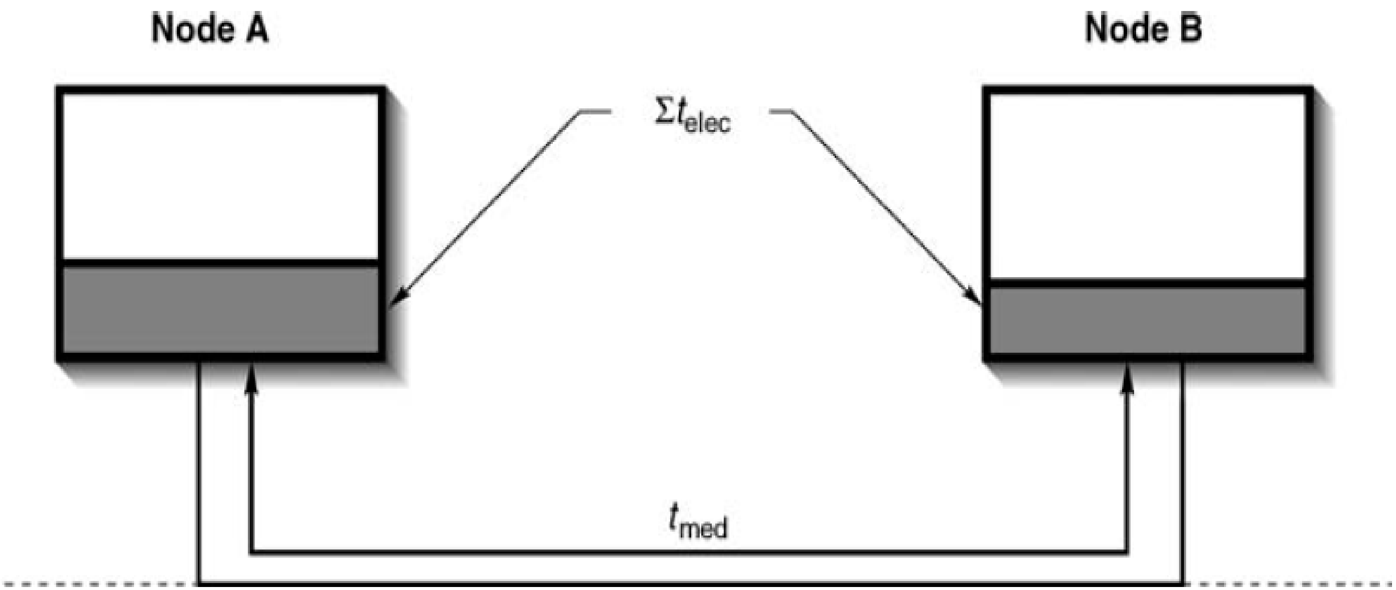
Propagation Segment Time

To precisely define the minimum period to be assigned to the propagation segment ($seg_{T\ prop}$), totality of the time contributions of all the elements in the network needs to be considered



Propagation Segment Time

- These parameters are frequently divided into two classes:
  - Those relating to the physical properties of the medium used (T_{med}) and
  - Those concerned with the electronic characteristics found in the network (T_{elec})
- Sum of these two components is T_{res}



$$T_{res} = T_{med} + T_{elec}$$

Data Rate and Bus Length

- Sender must be able to receive return data (e.g., arbitration)

- Therefore, propagation time segment' of the nominal bit time of a CAN network must therefore be greater than (or at least equal to) approximately twice T_{res}

$$seg_{T\ prop} \geq 2\ T_{res}$$

Data Rate and Bus Length

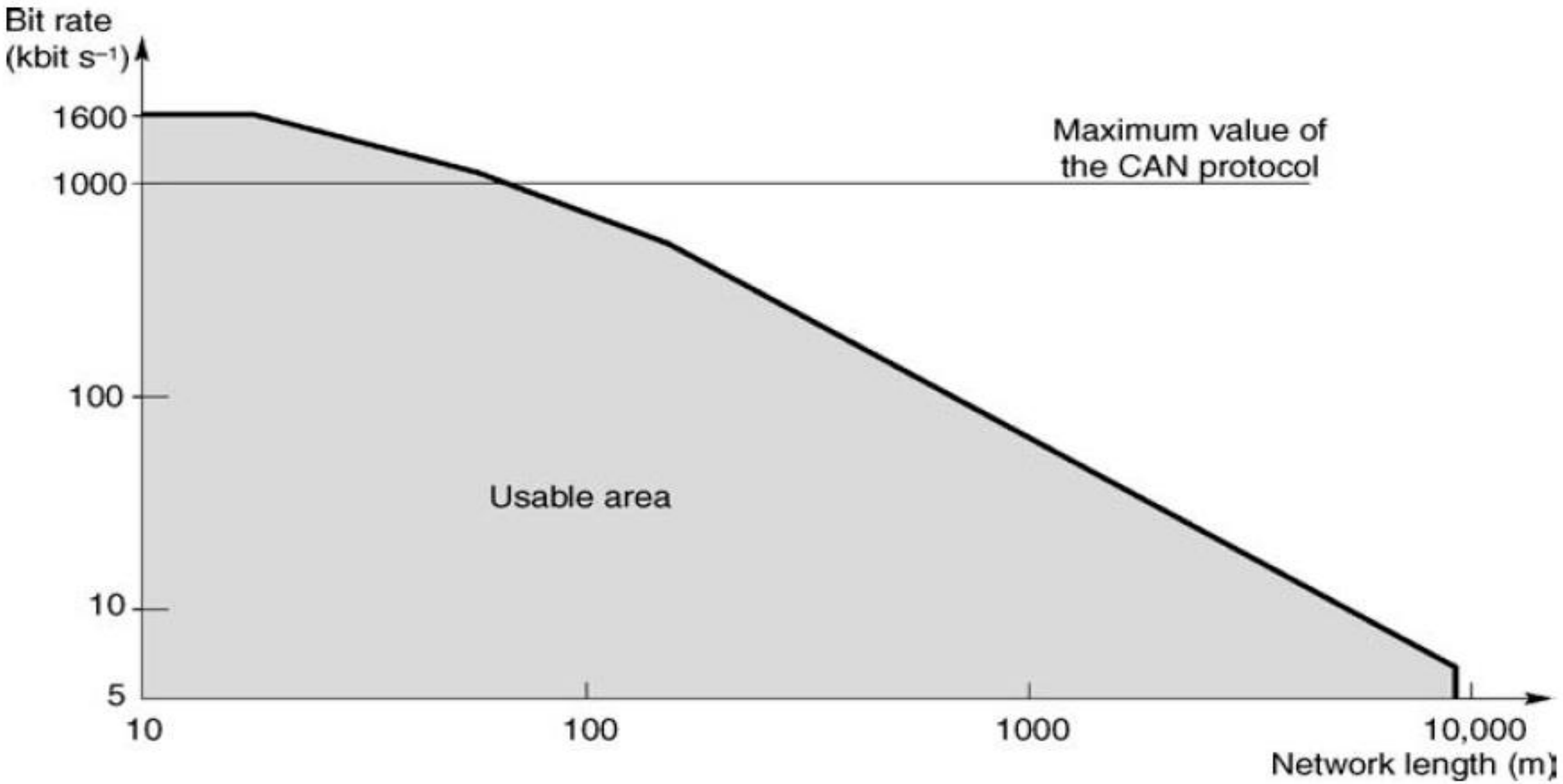
- For wired and optical medium propagation speed of the signal is equal to 0.2 m ns^{-1}
- Relation between length of the medium and bit rate is given by

$$L \leq (0.2\,\mathrm{m\,ns}^{-1})\left[\frac{x}{2\ \text{Baud rate}} - T_{\text{elec}}\right]$$

- Here, x is the percentage represented by propagation time segment

Data Rate and Bus Length

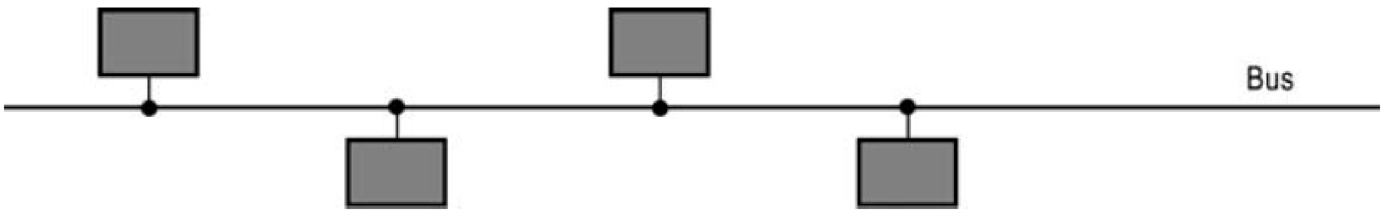
$T_{elec} = 100\,\text{ns},\ v_{prop} = 0.2\,\text{m ns}^{-1}$ (wire and optical medium), $x = 0.66 = 66\%$



Data Rate and Bus Length

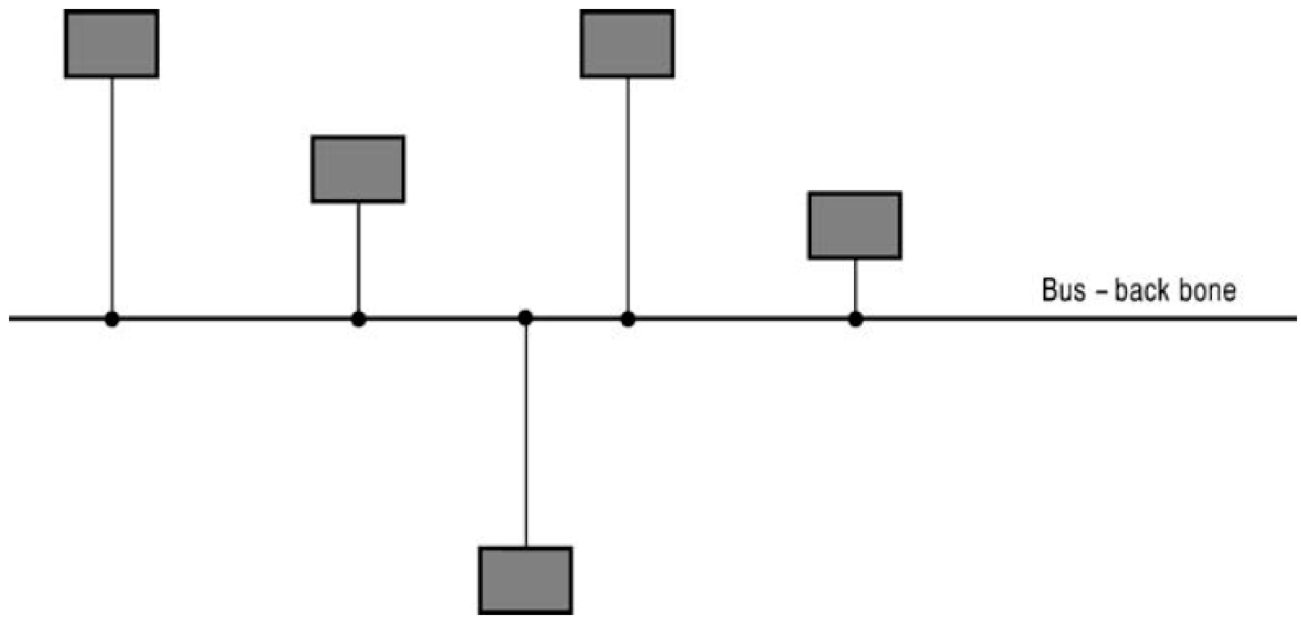
Bit rate	Maximum distance
1.6 Mbit s^{-1}	10 m
1 Mbit s^{-1}	40 m
500 kbit s^{-1}	130 m
250 kbit s^{-1}	270 m
125 kbit s^{-1}	530 m
100 kbit s^{-1}	620 m
50 kbit s^{-1}	1.3 km
20 kbit s^{-1}	3.3 km
10 kbit s^{-1}	6.7 km
5 kbit s^{-1}	10 km

Network Topology

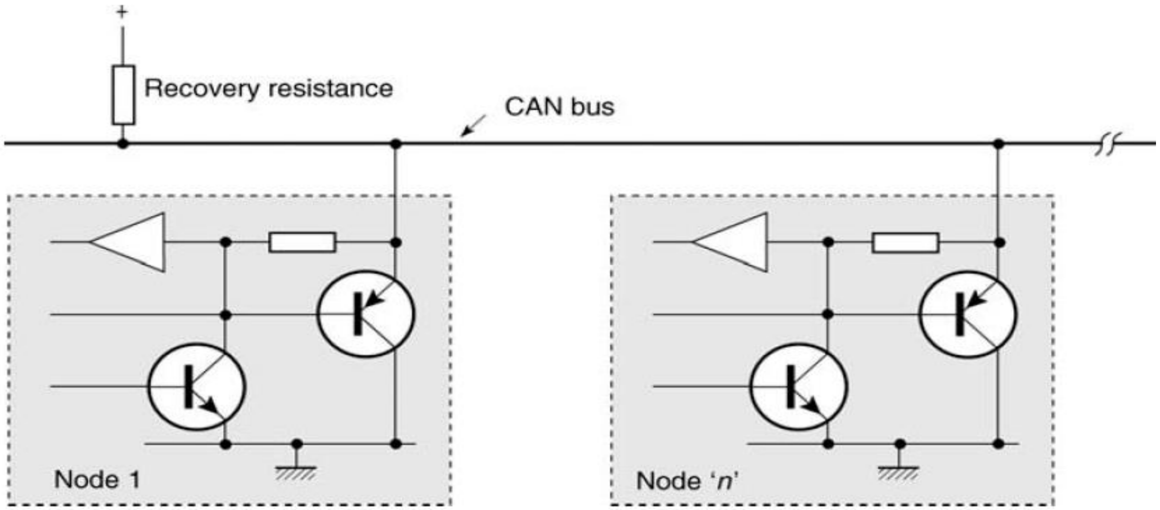


Bus

Backbone Bus

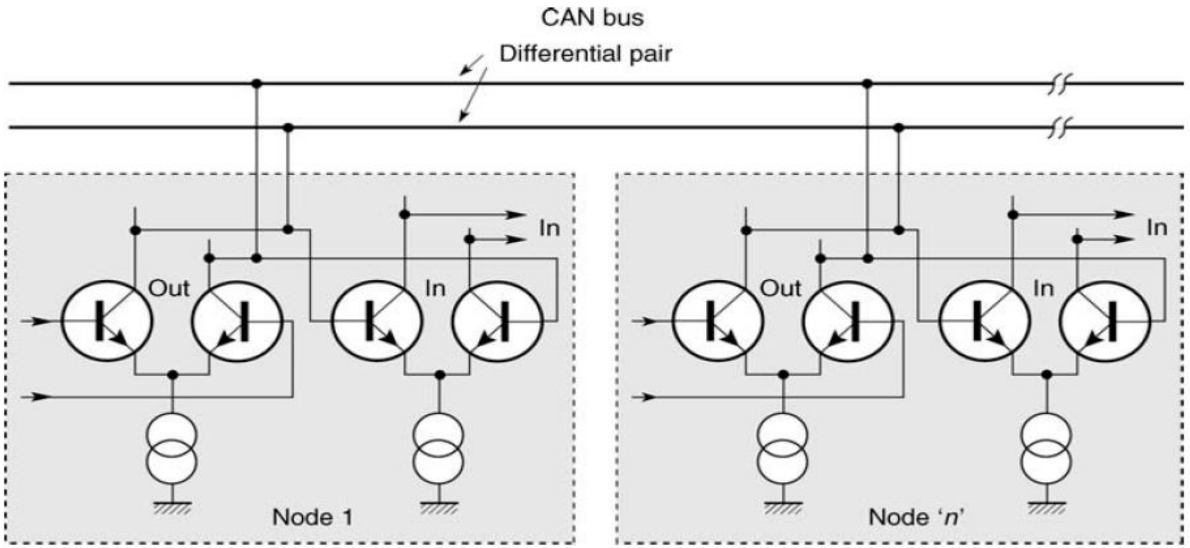


Physical Media



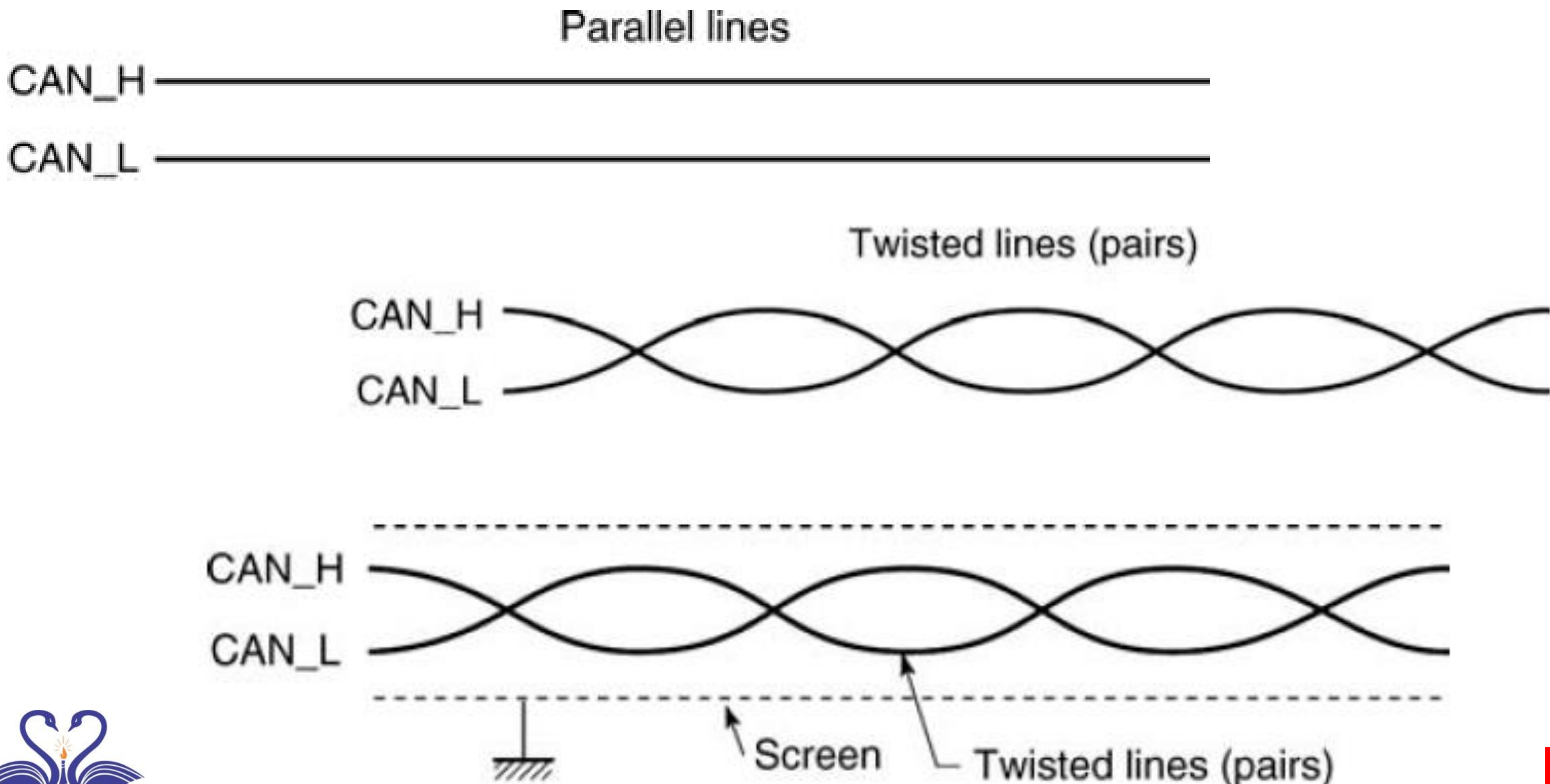
Single Wire Medium

Twin Wire Medium

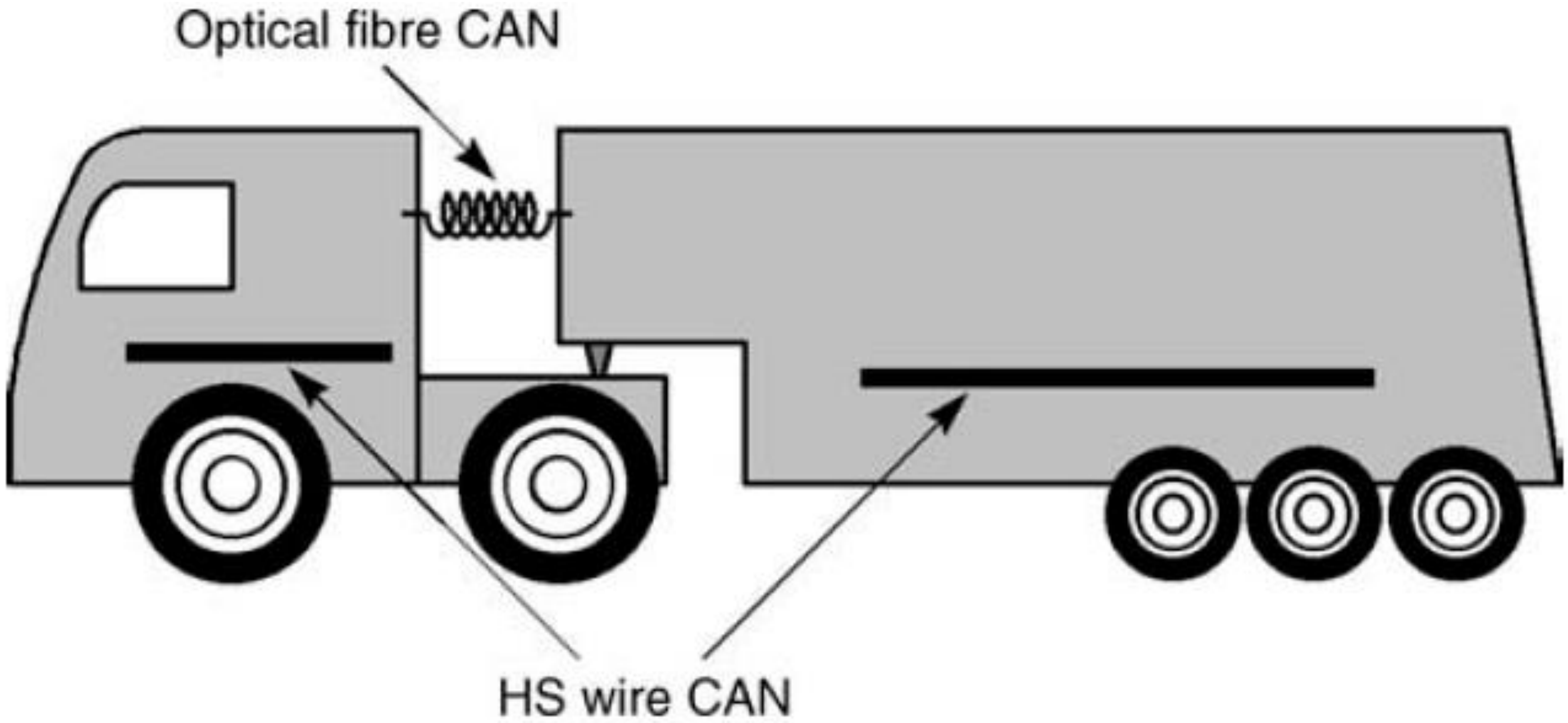


Physical Media

Twin wire medium

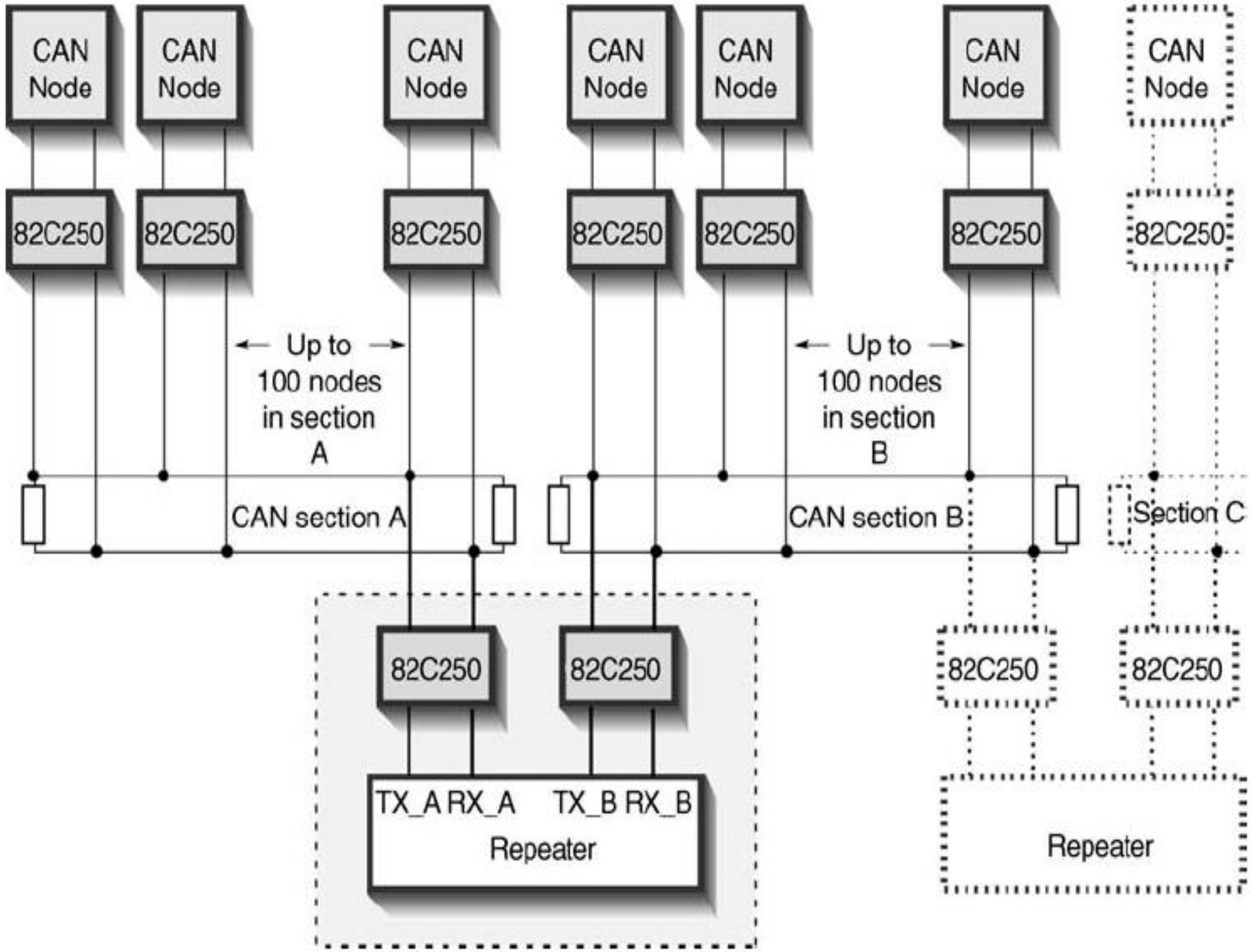


Optical

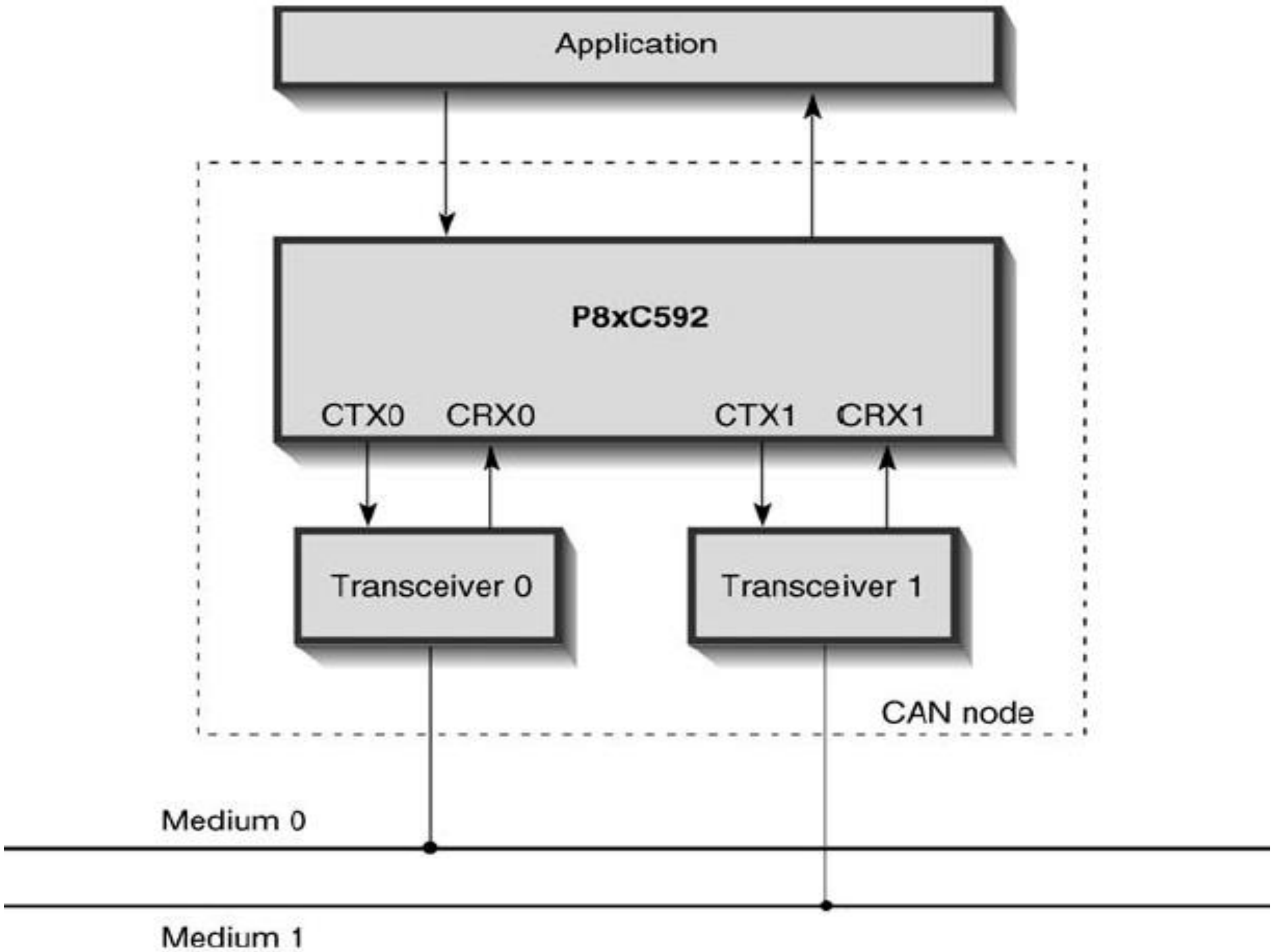


CAN optical fiber gateway between two CAN wire systems (a typical example is that of a 'tractor and trailer' unit)

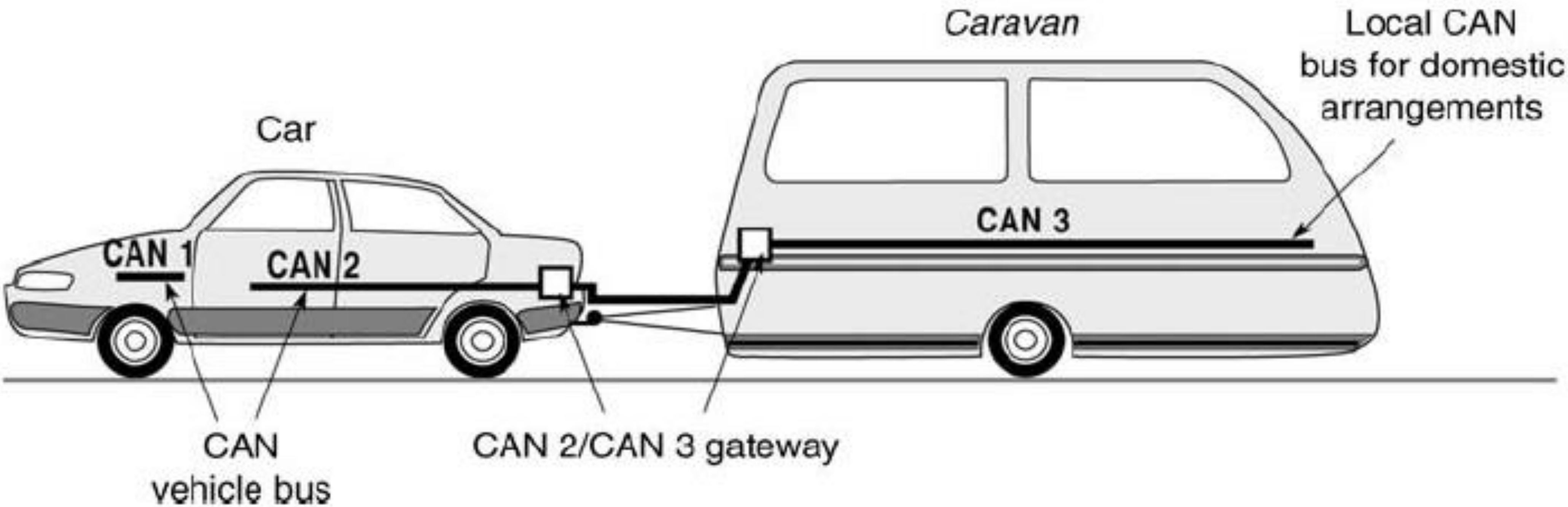
CAN Repeater



Medium-to-Medium Gateway



Medium-to-Medium Gateway



Time-Triggered Communication on CAN (TTCAN)

Introduction to TTCAN

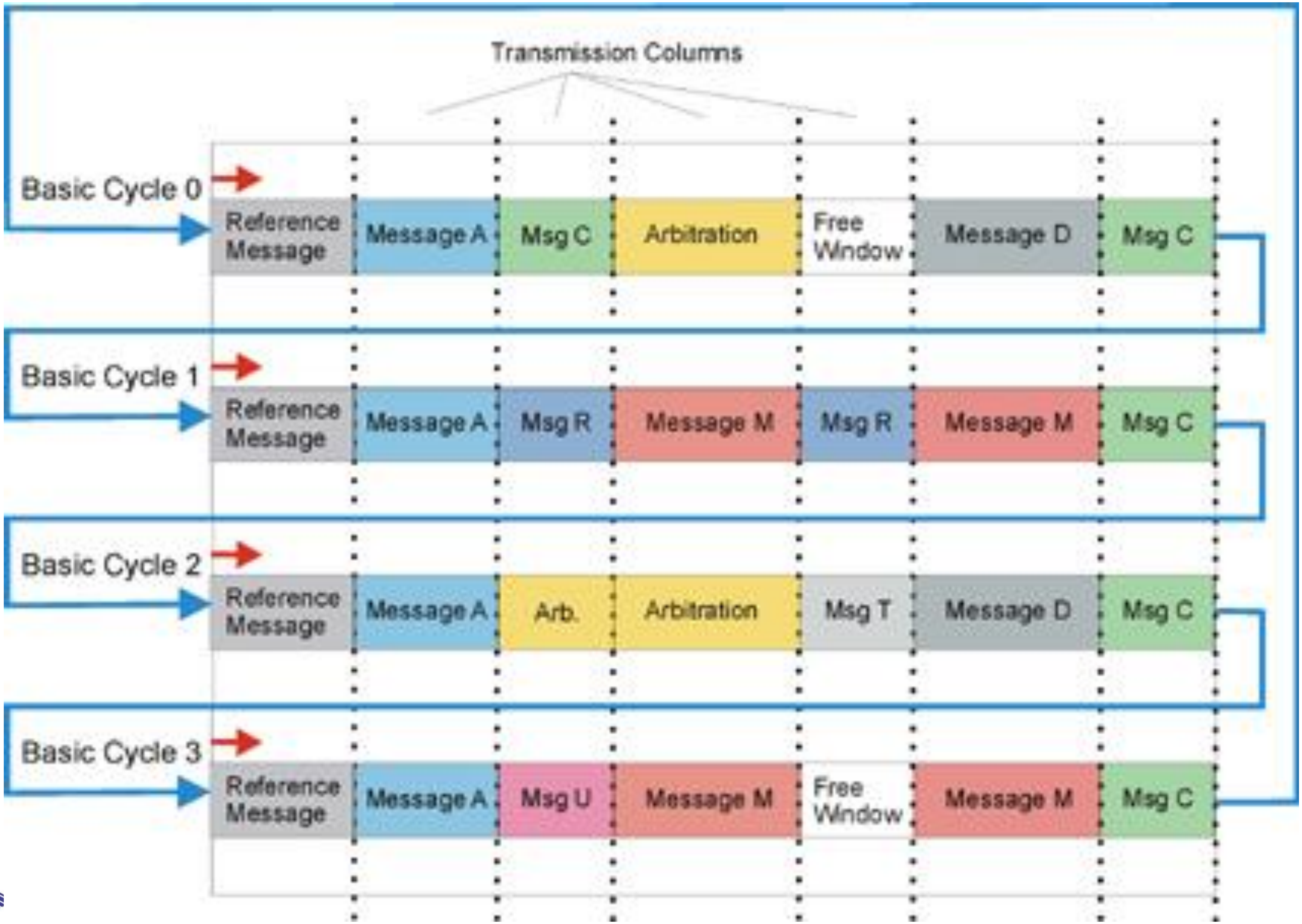
- TTCAN (time-triggered communication on CAN) protocol is a higher layer protocol on top of the CAN (Controller Area Network) data link layer

- Activities are triggered by the elapsing of time segments

- Time of message transmission is defined during the development of a system

- Ideal for applications in which the data traffic is of a periodic nature (safety critical systems: Status of brakes, airbags)

TTCAN Communication Cycle



Introduction to TTCAN

- Provides mechanisms to schedule CAN messages in a time-triggered way as well as in an event-triggered way
- Realized in software in a higher layer on top of CAN
- Reference message used for synchronization
- Reference message identified by nodes through its message identifier
- Individual TTCAN participants are configured to know when to send their frames after having received the reference frame

ज्ञानं विज्ञानं च भक्तिसहितं

Summary

- Controller Area Network (CAN) was initially created by German automotive system supplier Robert Bosch in the mid-1980s

- Characteristics of CAN protocol
  - Serial bus
  - Multi-master functionality
  - Bitwise arbitration
  - Message priority based bus access
  - Small messages (Maximum 8 Bytes)
  - Maximum data rate: 1 Mbps
  - Broadcast communication
  - Error detection and retransmission

- CAN hardware implementations cover the lower two layers of the OSI reference model

Summary

- CAN bus can have one of the two complementary logical values defined as
  - '0' as 'dominant' and
  - '1' as 'recessive'

- After 5 bits of identical value (either dominant or recessive), a supplementary bit having an intentionally opposite value is introduced

- Two types of synchronization
  - Hard synchronization
  - Resynchornization

- Physical layer (layer 1), PL:
  - Physical signaling sublayer, PLS (bit coding, bit timing, synchronization)
  - Physical medium attachment sublayer, PMA (characteristics of the command stages (drivers) and reception stages)
  - Medium dependent interface sublayer, MDI (connectors)

Reference

- Bosch, CAN (1991). *Specification version 2.0*. Published by Robert Bosch GmbH (September 1991)
- Bosch, CAN with Flexible Data Rate (2012). *Specification version 1.0.* Published by Robert Bosch GmbH (April 2012)
- Paret, D. (2007). *Multiplexed networks for embedded systems: CAN, LIN, Flexray, Safe-by-Wire...* John Wiley & Sons